

Transformers for Graph-based Recommender Systems: A Survey

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Abstract: Transformers are groundbreaking neural network architectures that have revolutionized natural language processing and have been adopted across a wide range of domains beyond text. Their ability to effectively handle sequential data has sparked growing interest in their application to recommender systems, which often involve sequential user-item interactions and contextual information that can naturally be represented as graphs. Thanks to their strength in capturing complex dependencies and patterns, transformers offer promising capabilities for enhancing recommender systems built on graph structures. In this survey, we present the first systematic overview of recent advances in graph-based recommender systems that leverage transformers. We provide a formal definition of graph-transformer-based recommender systems, propose a comprehensive taxonomy of existing approaches, and organize the relevant literature accordingly. Finally, we discuss current limitations and outline open challenges, pointing to directions for future research and development.

Keywords: Graphs, natural language processing (NLP), transformers, deep learning, recommender systems.

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1 Introduction

Recommender systems are widely known as tools for personalized information filtering aiming to estimate users' preferences on a set of items, in a variety of contexts, ranging from social^[1] to medical^[2] domains. To effectively address a recommendation task, it is essential to leverage appropriate representation models for both users and items in order to train machine learning methods that are designed to predict the users' preferences and provide recommendations. In recent years, there has been a surge of interest in the exploration of recommendation methods relying on observed user-item interactions^[3, 4]. Specifically, graph-based recommender systems (GRS) model user/item interactions through a graph structure, which is properly chosen to capture complex relational patterns and to leverage rich connectivity information inherent in graphs. The graph-based encoding has also proven effective in addressing common challenges in recommender systems, such as cold start problems, data sparsity and explainability issues^[5, 6].

Over the last few years, the predominant approach to GRS has been grounded in graph neural network (GNN)

models^[7], where user or item latent representations are learned through a message-passing scheme and then used to estimate users' preferences. GNNs have been widely used to effectively learn rich representations from various types of graphs used in recommender systems, such as user-item interaction graphs^[8, 9], social networks^[10], session-based graphs^[11] and heterogeneous graphs^[6]. GNNs are powerful, however, they are not always effective in building GRS^[12]. First, it is well-known that the receptive field of GNNs is constrained by the challenge of over-smoothing phenomena^[13], hence limiting their ability to capture long-term sequential information between users and items. Second, the message passing mechanism of GNNs can lead to strong biases and potential noise, since users may be engaged with products that do not align with their actual interests, due to over-recommendation of popular items^[14]. That is, users might relate to top-ranked items due to positional bias, rather than actual relevance, thus resulting in the formation of irrelevant connections within the user-item graphs^[15]. As a consequence, the message passing mechanism may be noisy and less effective^[16].

Motivations. The aforementioned limitations have driven researchers to explore the transformer neural learning framework^[17], a groundbreaking technology originating from the natural language processing (NLP) field. Specifically, transformers have become the dominant paradigm to build blocks constituting the state-of-the-art methods in NLP to successfully address a wide

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range of challenging tasks^[18], and their use is now expanding to graph data as well^[19, 20]. Their innovation lies in their ability to capture long-range dependencies and relationships between elements within a sequence, making them incredibly effective for tasks involving sequential data, such as language translation, text generation, sentiment analysis, and more. What makes transformers particularly powerful in NLP includes their self-attention (SA) mechanism, which allows them to weigh the importance of each word/token based on its relevance to other words/tokens in the sequence, enabling parallel processing and capturing complex relationships. Analogously, in graph representation learning, transformers can capture long-range dependencies between nodes without imposing prior structure on the input data. Indeed, recent findings have shown that transformers, when applied to the graph domain, frequently outperform GNNs on various benchmark datasets, including those for recommendation tasks^[21–24]. Indeed, GNNs might behave as low-pass filters^[25], smoothing the embeddings of connected user-item nodes. This homogenization blurs the distinctive signals that recommender systems need to rank items accurately.

With these premises, integrating transformer and GRS has the potential to boost personalized recommendation by leveraging user/item interaction patterns and the transformer's ability to handle long-term sequential data^[16, 26]. However, the fusion of transformer and graph-structured data in GRS scenarios has not been fully explored in the literature. Major challenges include building an effective graph topology-aware transformer^[27], dealing with computation and memory requirements of the SA module^[18], addressing the data sparsity inherent in recommender systems^[28], and exploiting the heterogeneity of edges and nodes^[26]. Recently, there have been developments of GRS to address some of these challenges based on a transformer approach. Nevertheless, there is a lack of reviews on GRS methods incorporating a transformer in the process of learning user preferences.

Contributions. In this survey, we fill the above mentioned gaps by providing a systematic overview of the recent advances in GRS approaches using transformers. Our key contributions can be summarized as follows: 1) We provide a formal definition of a graph transformer-based recommender system (GTRS) based on the current approaches proposed in the literature; 2) we provide a taxonomy to organize current works according to the manner in which graphs and transformers are integrated for modeling user preferences; 3) we review each work, highlighting key aspects, to what purpose transformer is used, and the corresponding performance of the reviewed models; 4) we discuss the current trends, limitations and future directions in the field.

Research methodology. This study reviews literature published from 2018, when transformers began receiv-

ing widespread attention, up to the present. We queried major scholarly databases, digital libraries and search engines for peer-reviewed publications from top-tier conferences and journals in recommender systems, data mining, and machine learning, thus ensuring the quality and relevance of the included literature. To this purpose, we used combinations of keywords related to the topic, including “graph-transformer-based recommendation”, “transformer recommender systems”, “graph neural network recommendation with transformer”, “graph transformer”, etc. Additionally, references of key papers and related surveys were cross-examined to capture any additional relevant studies.

The identified works span a variety of publication venues, learning strategies and domains, reflecting the growing interest in graph-based transformer recommenders. In particular, our selection of works regards those where a transformer neural network is used in conjunction with a graph-based component (such as a user-item interaction graph, knowledge graph, hypergraph, or social network) as part of the recommender system's design. Notably, we also considered approaches where the transformer and the graphs are both core components of the recommender architecture even if they do not directly interact with each other. By contrast, we excluded general background papers that do not introduce a new graph-transformer-based recommendation method.

Comparison with related surveys. To the best of our knowledge, this is the first survey on the field of GTRs. Other works have investigated recommender systems based on deep learning techniques^[29, 30]. On the one hand, the studies in ^[30, 31] explore the role of GNNs in recommender systems; on the other hand, Chen et al.^[32] focus on graph transformers in computer vision, whereas Min et al.^[33] explore graph transformers from an architectural design perspective, but with a broader scope. More recently, other works^[20, 34] explore the growing adoption and the foundations of graph transformers, yet without addressing their specific application to recommender systems. While existing surveys separately address GNN-based recommender systems or transformer applications in general graph-analysis tasks, a comprehensive overview dedicated explicitly to transformer-based approaches for graph recommendation is currently lacking. Addressing this gap is critical to guiding future research and facilitating the development of more effective recommendation models.

Our work bridges two distinct research areas: Transformers and GRS. It serves as a more focused counterpart to broader surveys that exclusively address GNN-based recommendation methods^[30, 31], deep learning approaches to graph representation learning^[29, 30], or graph transformers^[34]. By addressing the intersection of these domains, this survey provides a more targeted analysis of the specific challenges and opportunities arising from

both functional aspects (e.g., how graphs and transformers are integrated for recommendation) and architectural design choices (e.g., graph-aware adaptations of the self-attention mechanism). While advances limited to either GNN-based or transformer-based recommenders are beyond the scope of this work, the focus on recommendation tasks enables the development of a tailored taxonomy and facilitates deeper insights into the current landscape and future directions.

Unlike other studies addressing transformers for graphs, we aim to shed light on transformers within a GRS, hence we also consider works where the transformer is a core component but does not directly interact with a graph. Considering the widespread use of recommender systems across various real-world domains, and given the rapid adoption of transformer-based models specifically within graph-based recommendation scenarios, we believe this survey on GTRS is particularly timely.

2 Preliminaries

In this section, we briefly describe the vanilla transformer neural framework^[17]. Next, we elaborate on the concept of GTRS and the evaluation measures typically used by the works analyzed in this survey.

Transformer. The vanilla transformer architecture consists of stacked encoder-decoder blocks^[17]. The encoder is composed of several identical layers, each containing two primary sublayers: a self-attention mechanism (SA) and a position-wise feed-forward neural network (FFN). For building deeper models, a residual connection layer^[35], followed by a layer normalization (LN)^[36] module, is also employed.

Before feeding the input to the first transformer layer, a positional encoding is added to the input embeddings to inject information about the relative positions of tokens within the sequence. Since the transformer architecture does not inherently capture positional information, positional encoding explicitly encodes the position of each entity (e.g., token or node) using fixed sinusoidal functions or learned embeddings. Formally, let $\mathbf{H} = [\mathbf{h}_1^T, \dots, \mathbf{h}_n^T] \in \mathbb{R}^{n \times d}$ be the input of the transformer, where n is the number of entities, d is the hidden dimensionality, and each $\mathbf{h}_i \in \mathbb{R}^d$ represents the embedding (column) vector for the i -th entity. The positional encoding matrix $\mathbf{P} \in \mathbb{R}^{n \times d}$ provides position-dependent signals, e.g., computed using fixed sinusoidal or learned functions^[37]. Typically, $\mathbf{H}$ and $\mathbf{P}$ are added element-wise, thus obtaining $\mathbf{H}' = \mathbf{H} + \mathbf{P}$, which merges semantic and positional information. The position-aware representation $\mathbf{H}'$ is then used by the subsequent transformer layer. Specifically, the SA module transforms the input with the projection matrices $\mathbf{W}_Q \in \mathbb{R}^{d \times d_k}$, $\mathbf{W}_K \in \mathbb{R}^{d \times d_k}$, and $\mathbf{W}_V \in \mathbb{R}^{d \times d_v}$, generating the corresponding representations $\mathbf{Q}, \mathbf{K}, \mathbf{V}$ as follows:

$$\begin{aligned} \mathbf{Q} &= \mathbf{H}'\mathbf{W}_Q, \quad \mathbf{K} = \mathbf{H}'\mathbf{W}_K, \quad \mathbf{V} = \mathbf{H}'\mathbf{W}_V, \\ \mathbf{A} &= \frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}, \quad \text{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}(\mathbf{A})\mathbf{V} \end{aligned} \quad (1)$$

where $\mathbf{A}$ is the matrix representing the similarity between queries and keys, and the term d_k is a scaling factor. The softmax function ensures that each row of $\mathbf{A}$ sums to 1, interpreting attention scores as probabilities over entities. Note that, to incorporate biases or constraints based on domain priors within the attention computation, the attention matrix $\mathbf{A}$ can be multiplied element-wise by a mask M that has the same shape as $\mathbf{A}$ ^[18, 20].

The transformer employs multiple self-attention mechanisms, referred to as multi-head self-attention (MHSA). Specifically, MHSA consists of T distinct attention mechanisms, each termed a head and indexed by t . Each head captures different types of relationships from separate representation subspaces, allowing the encoder to simultaneously attend to information from multiple perspectives:

$$\begin{aligned} \mathbf{Q}^{(t)} &= \mathbf{H}'\mathbf{W}_Q^{(t)}, \quad \mathbf{K}^{(t)} = \mathbf{H}'\mathbf{W}_K^{(t)}, \quad \mathbf{V}^{(t)} = \mathbf{H}'\mathbf{W}_V^{(t)} \\ \text{Attn}^{(t)} &= \text{Attn}(\mathbf{Q}^{(t)}, \mathbf{K}^{(t)}, \mathbf{V}^{(t)}) \\ \text{MultiHead}(\mathbf{H}') &= \text{concat}(\text{Attn}^{(1)}, \dots, \text{Attn}^{(T)})\mathbf{W}^O \end{aligned} \quad (2)$$

where T is the number of attention heads, $\mathbf{W}_Q^{(t)} \in \mathbb{R}^{d \times d_k}$, $\mathbf{W}_K^{(t)} \in \mathbb{R}^{d \times d_k}$, and $\mathbf{W}_V^{(t)} \in \mathbb{R}^{d \times d_v}$ are the projection matrices for queries, keys, and values, respectively, specific to each head t ; $\mathbf{W}^{(O)} \in \mathbb{R}^{Qd_v \times d}$ is the output projection matrix, which combines the outputs from all heads.

Each transformer block also employs a residual connection followed by a layer normalization step, defined as $\mathbf{H}'' = \text{LN}(\mathbf{H}' + \text{MultiHead}(\mathbf{H}'))$, where LN denotes the layer normalization, stabilizing training by normalizing the layer outputs across features. In the last few years, the normalization layer has received increasing attention for improving the performance of transformers^[38]. Following this layer, the output is processed by a position-wise FFN:

$$\text{FFN}(\mathbf{H}'') = \max(0, \mathbf{H}''\mathbf{W}_1 + \mathbf{b}_1)\mathbf{W}_2 + \mathbf{b}_2 \quad (3)$$

where $\mathbf{W}$ s are trainable matrices, and $\mathbf{b}$ s are bias terms. The FFN independently applies the same transformation to each position.

Finally, another residual connection and normalization step are applied, producing the final output of a transformer block: $\mathbf{H}_{\text{out}} = \text{LN}(\mathbf{H}'' + \text{FFN}(\mathbf{H}''))$. These residual connections facilitate gradient flow, enabling effective training of deeper architectures.

The decoder follows a similar structure, but it em-

employs a masked multi-head self-attention mechanism to ensure that, while predicting the current word, it does not access future tokens, thereby functioning in an auto-regressive fashion. Specifically, the SA module can only attend to key-value pairs up to and including the current position.

Graph-transformer-based recommendation. The overview of a GTRS, based on the works reviewed in this survey, is shown in Fig. 1. Unlike GNNs, which naturally incorporate an inductive bias based on locality and permutation invariance, transformers do not assume any inherent structure in the data. Therefore, it is necessary to design a GTRS as an architecture that explicitly integrates both the transformer mechanism and the structural prior of graphs into its computation.

Formally, let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a general graph representation, where $\mathcal{V}$ is the set of nodes (with $|\mathcal{V}| = n$) representing entities, and $\mathcal{E}$ is the set of edges representing relationships or interactions among these entities. The graph $\mathcal{G}$ can be any form, including but not limited to simple graphs, directed graphs, weighted graphs, heterogeneous graphs, or hypergraphs, thus allowing for a very general representation of relationships. In a typical recommender system, users $\mathcal{U}$ (e.g., individual profiles or customer accounts) and items $\mathcal{I}$ (e.g., products, services, or media) are included among the entities $\mathcal{V}$, since they represent specific categories of nodes that interact with each other. Each entity $v \in \mathcal{V}$ might be associated with external side information. Let $\mathbf{X} \in \mathbb{R}^{n \times f}$ be a matrix of feature vectors where each row $\mathbf{x}_v \in \mathbb{R}^f$ is an f -dimensional feature vector modeling structural features, attribute information, textual descriptions, or any side information relevant to node $v \in \mathcal{V}$.

We also denote with $\mathcal{T}$, a transformer-based model, namely, a neural architecture characterized by trainable parameters $\Theta_{\mathcal{T}}$ and employing the MHSA learning entity representations, as detailed in Section 2. Specifically, $\mathcal{T}$ operates on the transformed input embeddings, $\mathcal{R}(\mathcal{G}, \mathbf{X})$, which may encode graph-aware information through the function $\mathcal{R}$. Additionally, it can incorporate structural priors, given by the function $\mathcal{M}(\mathcal{G})$, thereby

potentially modifying how the attention scores are computed ((1) and (2)). Based on the above, we define a GTRS through the sequential application of (4) and (5). The former is defined as follows:

$$\mathbf{Z} = \mathcal{F}\left(\mathcal{T}(\mathcal{R}(\mathbf{X}, \mathcal{G}; \Theta_{\mathcal{R}}), \mathcal{M}(\mathcal{G}; \Theta_{\mathcal{M}}); \Theta_{\mathcal{T}}), \mathcal{G}, \mathbf{X}; \Theta_{\mathcal{F}}\right) \quad (4)$$

where $\Theta = \{\Theta_{\mathcal{R}}, \Theta_{\mathcal{M}}, \Theta_{\mathcal{T}}, \Theta_{\mathcal{F}}\}$ is the full set of learnable parameters. $\mathcal{R}(\mathbf{X}, \mathcal{G}; \Theta_{\mathcal{R}})$ transforms $\mathbf{X}$ into (possibly graph-enriched) node embeddings. Note that $\mathcal{R}(\cdot; \Theta_{\mathcal{R}})$ might be an identity mapping, thus providing graph-agnostic features with shape (n, d) . It typically produces a matrix $\mathbf{H}' \in \mathbb{R}^{n \times d}$. $\mathcal{M}(\cdot; \Theta_{\mathcal{M}})$ produces a structural prior or mask that is integrated within (1) or (2) for attention based on $\mathcal{G}$ – for instance a matrix $\mathbb{R}^{n \times n}$ or another tensor structure suitable for the integration into the attention mechanism. This prior constrains the transformer's attention. If $\mathcal{M}$ ignores $\mathcal{G}$, it might produce a trivial full mask attention. $\mathcal{T}(\cdot; \Theta_{\mathcal{T}})$ is the transformer module (potentially comprising multiple layers) that receives node embeddings from $\mathcal{R}(\cdot; \Theta_{\mathcal{R}})$ and a structural prior from $\mathcal{M}(\cdot; \Theta_{\mathcal{M}})$. $\mathcal{F}(\cdot; \Theta_{\mathcal{F}})$ is the final neural block that processes the transformer's output – possibly in combination with other signals – to produce the final node representations $\mathbf{Z}$. Note that $\mathbf{Z} \in \mathbb{R}^{n \times d'}$ includes the representation of all the entities in $\mathcal{V}$, such as $\mathcal{U}$ or $\mathcal{I}$. The recommendations are then computed by defining a scoring function (aka recommendation model) g that predicts the preference score $r_{u,i}$ as follows:

$$r_{u,i} = g(\mathbf{z}_u, \mathbf{z}_i; \Theta_g) \quad (5)$$

where $\mathbf{z}_u, \mathbf{z}_i \in \mathbf{Z}$ are user and item embeddings (rows of the matrix $\mathbf{Z}$) produced by $\mathcal{F}$, and Θ_g are parameters of the recommendation function g . The scoring function g is typically based on the inner product between the input embeddings, e.g., $\mathbf{z}_u^T \mathbf{z}_i$.

Therefore, $\mathcal{F}$ produces the final embeddings, integrating graph-based and transformer-extracted information, while g uses these embeddings to produce the recom-

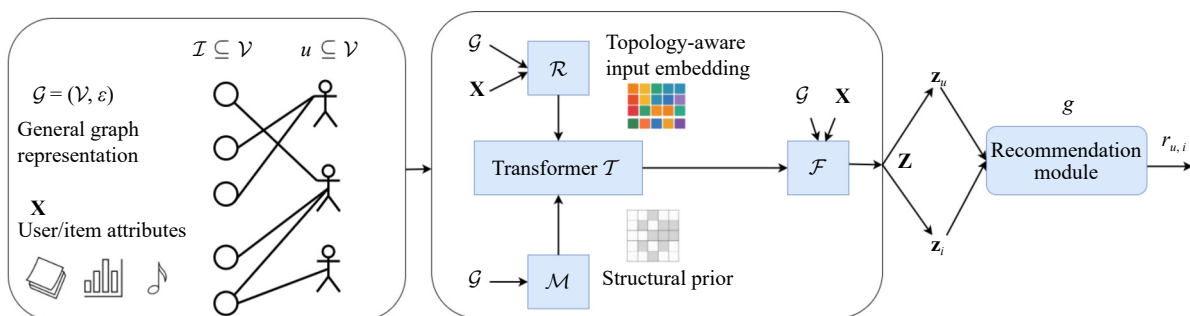


Fig. 1 Conceptual overview of a GTRS, derived from the analysis of existing works. The GTRS integrates user/item attributes and a general interaction graph into a transformer architecture, conditioned by the topology-aware embeddings and structural priors, to learn node representations. Final embeddings are used to predict user-item preference scores via a recommendation module. (Colored figures are available in the online version at <https://link.springer.com/journal/11633>)

mentation scores $r_{u,i}$. The synergy of $\mathcal{F}$ and g – together with the components $\mathcal{R}$, $\mathcal{M}$, and $\mathcal{T}$ – defines a GTRS.

We emphasize that, for a model to qualify as a GTRS, the output of at least one of the functions $\mathcal{R}$, $\mathcal{M}$, or $\mathcal{F}$ must depend non-trivially on the graph $\mathcal{G}$. Formally

$$\exists \phi \in \{\mathcal{R}, \mathcal{M}, \mathcal{F}\} : \frac{\partial \phi(\mathbf{X}, \mathcal{G}; \Theta_\phi)}{\partial \mathcal{G}} \neq 0 \quad (6)$$

where $\partial/\partial \mathcal{G}$ denotes the partial derivative with respect to a real-valued tensor representation of the graph (e.g., its adjacency matrix possibly concatenated with node or edge attributes, or a learned dense embedding derived from $\mathcal{G}$). Note that some modules, such as $\mathcal{M}$, may not consume node features, so the argument $\mathbf{X}$ can be omitted without loss of generality.

To summarize, the general formulation in (4) allows the modeling of different types of GTRS, as we will show in Section 3.

Evaluation measures in recommender systems.

Evaluating recommender systems often requires restricting relevant items to the k most relevant items^[39]. Throughout this survey, we use the $@k$ notation for common evaluation measures in recommendation, where k is the rank cutoff for the list of recommended items. The mean reciprocal rank (MRR $@k$)^[40] assesses the quality of a ranked list by focusing on the rank of the first relevant item found within the top k recommendations. It is calculated as the average of the reciprocal ranks across all users (or queries) to emphasize the recommendation of at least one relevant item as early as possible in the ranking. Another ranking-based method is the normalized discounted cumulative gain (NDCG $@k$)^[41], which measures the quality of the ranking by considering the positions and relevance scores of all relevant items within the top k recommendations. It is the normalized version of discounted cumulative gain (DCG)^[42], which aggregates the relevance scores (gains) of items, applying a logarithmic dis-

count based on their rank position (items ranked higher contribute more). The Hit rate (H $@k$) indicates whether at least one relevant item is included among the top k recommendations for a user^[42].

Recall $@k$ ^[42] measures the proportion of all relevant items for a user that are successfully included in the top k recommendations. It quantifies the system's ability to retrieve relevant items (coverage), irrespective of their specific order within the top k . Higher recall indicates that the system finds a larger fraction of the items that the user actually likes. Precision $@k$ ^[42] is the proportion of the top k recommended items that are relevant. It reflects recommendation accuracy at cutoff k . The balance between Precision $@k$ and Recall $@k$ can be summarized by F1-score $@k$, which corresponds to their harmonic mean. It provides a single score that balances the recommender's ability to retrieve many relevant items (recall) with its accuracy in the top k (precision). A higher F1-score $@k$ indicates a better trade-off between precision and recall.

In cases where the analyzed studies do not provide evaluations using metrics $@k$, we resort to using area under the receiver operating characteristic curve (AUC), mean absolute error (MAE) and root mean square error (RMSE) as alternative measures. The area under the ROC curve (AUC-ROC)^[43] measures the likelihood that a random relevant item is ranked higher than a random irrelevant item. In recommendation, AUC is often used to evaluate the model's overall ranking quality across all recommendation thresholds. MAE^[44] measures the average magnitude of prediction errors, regardless of their direction, thus quantifying how closely predictions match actual user ratings. RMSE^[45] further penalizes larger errors by squaring them before averaging, making it particularly sensitive to outliers.

3 Proposed taxonomy of approaches

We propose a taxonomy of GTRS approaches based

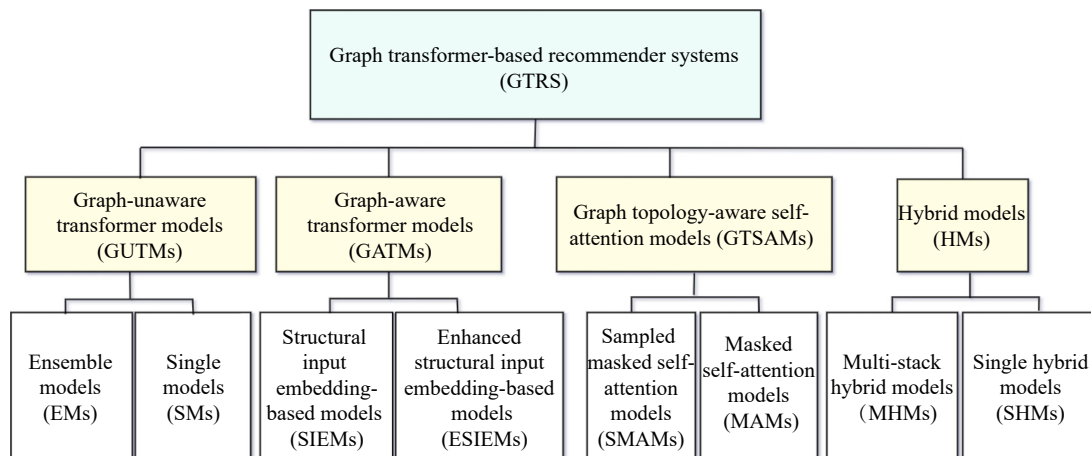


Fig. 2 Overview of the proposed GTRS taxonomy (Colored figures are available in the online version at <https://link.springer.com/journal/11633>)

on 4 main functional categories (i.e., the 1st level in the taxonomy) and 6 architectural subcategories (i.e., the 2nd level in the taxonomy), as sketched in Fig. 2. The first-level categorization is functional, as it reflects the role that graph information plays within the transformer core. Specifically, it captures the presence or absence of two types of graph-induced bias: 1) node-level representation priors injected via $\mathcal{R}$, and 2) structural priors enforced through $\mathcal{M}$. Fig. 3 illustrates a decision tree in which the root and internal nodes correspond to decision points, and the leaves represent the final categories of our taxonomy. For example, the question “Does $\mathcal{R}$ depend on prior information from $\mathcal{G}$?” can be assessed by verifying

$$\frac{\partial}{\partial \mathcal{G}} \mathcal{R}(\mathbf{X}, \mathcal{G}; \Theta_{\mathcal{R}}) \neq 0 \quad (7)$$

where, once again, the derivative $\frac{\partial}{\partial \mathcal{G}}$ is taken with respect to a real-valued tensor representation of the graph. A zero derivative implies that $\mathcal{R}$ is graph-agnostic; a non-zero value indicates that $\mathcal{R}$ exploits the graph structure. Based on such assessments for $\mathcal{R}$, $\mathcal{M}$, and $\mathcal{F}$, a mutually exclusive category is then assigned within the taxonomy.

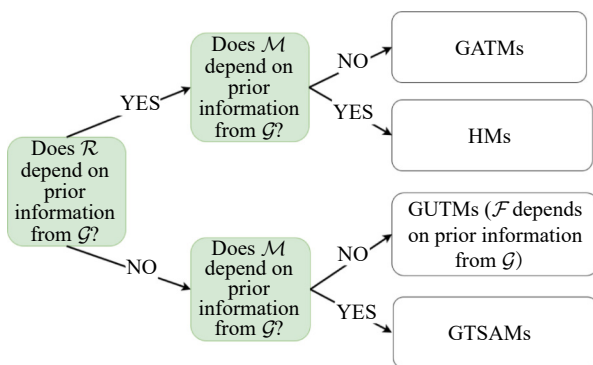


Fig. 3 Decision tree of the proposed taxonomy. Green and white colors indicate decision and leaf nodes, respectively. (Colored figures are available in the online version at <https://link.springer.com/journal/11633>)

Note that a system does not qualify as a GTRS if none of the functions $\mathcal{R}$, $\mathcal{M}$, or $\mathcal{F}$ utilizes the graph. Moreover, since the first level of the taxonomy is function-oriented, novel approaches integrating GRS and transformers can be seamlessly mapped onto the existing framework without requiring any modification. The second-level categorization is instead architectural, since it specifies the specific design choices made in implementing $\mathcal{R}$, $\mathcal{M}$ or $\mathcal{F}$.

GUTMs consider a transformer relying solely on content-based information about users and items within a GRS. In GUTMs, the transformer model $\mathcal{T}$ itself does not explicitly encode any graph structural information via its primary input sequence or its SA mechanism. Specifically, a model is GUTM if neither $\mathcal{R}(\cdot, \cdot; \Theta_{\mathcal{R}})$ nor $\mathcal{M}(\cdot; \Theta_{\mathcal{M}})$

use $\mathcal{G}$ in their computation. In this case, the former is a function of $\mathbf{X}$, while the latter returns the full attention mask, i.e., $\mathbf{1}\mathbf{1}^T$, with $\mathbf{1}$ being the $n \times 1$ vector of ones. Nevertheless, $\mathcal{G}$ and $\mathbf{X}$ are still used by another aggregator function encapsulated within $\mathcal{F}$, but not directly within the core transformer, i.e., $\mathcal{F} = (\cdot; \mathcal{G}, \mathbf{X}; \Theta_{\mathcal{F}})$ – for instance, some approaches leverage self-supervised signals extracted from the graph during training. GUTMs are further categorized into single model (SM) and ensemble model (EM). In SMs, the transformer is employed within a serial architecture, while in EMs the transformer is employed within a parallel architecture, where the output of different models is combined for computing the final recommendation. Note that (4) can handle both cases through proper decompositions of the function $\mathcal{F}(\cdot, \cdot, \cdot; \Theta_{\mathcal{F}})$. We emphasize that a GUTM is graph-unaware at the transformer level: The SA layers operate exclusively on content such as text or images. Structural information is injected only downstream, through $\mathcal{F}$. This is formally described as follows:

$$\frac{\partial \mathcal{R}(\mathbf{X}, \mathcal{G}; \Theta_{\mathcal{R}})}{\partial \mathcal{G}} = 0 \wedge \frac{\partial \mathcal{M}(\mathcal{G}; \Theta_{\mathcal{M}})}{\partial \mathcal{G}} = 0 \wedge \frac{\partial \mathcal{F}(\mathbf{X}, \mathcal{G}; \Theta_{\mathcal{F}})}{\partial \mathcal{G}} \neq 0. \quad (8)$$

GATMs refer to a transformer leveraging graph topological information in the form of input node latent representations (aka embeddings). These embeddings are given as input to a transformer that learns enhanced user/item representations useful for recommendation. Formally, in GATMs, the graph structure $\mathcal{G}$ is encoded explicitly into node embeddings before being input to $\mathcal{T}$. Thus, the function $\mathcal{R}(\mathcal{G}, \mathbf{X}; \Theta_{\mathcal{R}})$ is employed to compute node embeddings based on graph structural information, while $\mathcal{M}$ is the full-attention mask, i.e., $\mathcal{M}(\mathcal{G}; \Theta_{\mathcal{M}}) = \mathbf{1}\mathbf{1}^T$. GATMs further specialize into two sub-categories: SIEMs, which employ the structure embedding to directly initialize the transformer, and ESIEMs, where the embedding is integrated with other encoded information for enhancing the model learning capabilities.

GTSAMs consider topological information that is exploited solely by injecting graph priors into the self-SA block. In GTSAMs, the function $\mathcal{M}(\cdot; \Theta_{\mathcal{M}})$ explicitly uses $\mathcal{G}$ – for instance, an adjacency-based mask or a neighbor weighting mechanism – while the input node features computed through $\mathcal{R}(\cdot, \cdot; \Theta_{\mathcal{R}})$ are purely content-based, meaning that $\mathcal{G}$ is neglected. In this case, two categories arise: the first one is MAMs in which a graph masking mechanism enforces nodes to attend over a subset of positions, whereas in SMAMs the masking mechanism is further enhanced with a sampling procedure.

HMs include approaches that are both GATM and GTSAMs, i.e., the transformer is fed with graph-aware embeddings and its SA mechanism injects graph priors. Therefore, both the functions $\mathcal{R}$ and $\mathcal{M}$ are simultaneously employed to inject graph information into the

transformer for the final recommendation. Because HMs admit graph priors along both node-wise structural embeddings, embedded via $\mathcal{R}$, and pair-wise routing constraints, enforced by $\mathcal{M}$, they capture automorphism-invariant node semantics and topology-aware message paths within the same hypothesis space. This dual conditioning, at least in principle, endows them with a strictly greater expressive capacity than either GATMs or GT-SAMs. As shown in different works^[46–48], leveraging both node-wise structural embeddings and routing constraints into the SA mechanism enhances representation power of graph transformer.

In HMs, two macro categories arise. MHMs employ two or more transformer stacks: At least one stack applies the routing mask $\mathcal{M}$, another ingests graph-aware embeddings produced by $\mathcal{R}$ (optionally generated by a GNN or auxiliary transformer). SHMs inject $\mathcal{R}$ and $\mathcal{M}$ within the same transformer block simultaneously.

Note that a GTRS can also inject graph information into the transformer via a positional encoding matrix computed from the graph (e.g., Laplacian positional encodings), without relying on an explicit structural embedding vector. We emphasize that $\mathcal{R}$ is considered to be active in this case as well, and the corresponding GTRS is therefore classified as a GATM – or as an HM if it additionally incorporates priors through $\mathcal{M}$ – but never as a GUTM.

In Table 1, we provide a summary of the works selected in this survey where, besides organizing them according to the proposed taxonomy (sub)categories, we report information about the recommendation task, the type of graph modeled to solve the task (i.e., bipartite, homogeneous, heterogeneous, hypergraph), the transformer architecture (e.g., encoder), the data used for evaluation, and the source code availability. Note that, concerning the tasks, we distinguish between general recommendation tasks and those focusing on modeling specific aspects: Sequential recommendation predicts the next user interaction, while session recommendation focuses on user actions within a session; click-through rate (CTR) prediction is about predicting user click likelihood; multimodal recommendation employs different types of user/item side information; multi-behavior recommendation predicts user preferences across different types of interactions; point of interest (POI) recommendation forecasts user future movements; conversational recommendation suggests items through language-based interactions; other domain-specific tasks, including paper, product, medication, and recipe recommendation. Web APIs recommendation is the task of recommending the APIs that a mashup – a web-integrated service combining multiple web APIs – is most likely to invoke.

In Section 4, we elaborate on each of the works in Table 1; note that we will indicate the used dataset and,

when available, we will mention main evaluation results achieved by an approach according to standard criteria used in recommender systems, such as MRR, NDCG, F1-score and AUC.

4 Approaches

4.1 Single models

Graph enhanced representation learning (GERL)^[49] deals with the news recommendation task, i.e., estimating user preferences for news. Given the inherent clarity of news titles, GERL first utilizes a transformer with a single layer of MHSA for learning semantic representations of news and users, and then it attentively aggregates the transformer encoded content of a user's clicked news. By exploiting a bipartite graph built upon user clicks, and the user/news representations, a graph attention network (GAT)^[96] is used to learn enhanced user/news representation for producing the final news recommendation. GERL achieves NDCG@10 of 0.31 on MSN News data.¹

Temporal graph transformer (TGT)^[50] predicts the next user actions within a specific behavior, considering other behaviors as context. Given identity embedding enhanced with temporal information as input token, a MHSA mechanism computes the item representations for each behavior type, across different items. A behavior-aware message passing scheme, with messages propagating from item to user side, generates the user embedding based on a specific behavior type. TGT yields NDCG@10 of 0.26 and 0.33 on two e-commerce datasets: Taobao and IJCAI.

Knowledge graph and transformer encoder for personalized paper recommendation (KGTE)^[52] employs bidirectional encoder representations from transformers (BERT) for learning author and paper embeddings from their attribute information. These embeddings are then used for constructing a knowledge graph, which includes author, keyword, paper-doi and cite-doi as entities, and different relations among them (e.g., paper-doi include keyword). The final recommendation is obtained by applying a three-layer RippleNet^[97] based on the heterogeneous information obtained by fusing meta-paths. KGTE yields NDCG@10 of 0.66 on two academic datasets, that is, DBLP and Aminer.

Unified multi-modal graph transformer (UGT)^[51] addresses the multi-modal recommendation task by jointly leveraging a feature extraction module and a user-item graph. Specifically, it leverages a multi-way transformer that tokenizes images into patches and text into word tokens, and then applies a shared multi-head attention to align the visual and textual tokens in a single SA block. However, in the feed-forward stage, the network splits by modality (modality experts): Image tokens pass through

¹ <https://www.msn.com/en-us/news>.

Table 1 Summary of the selected works in this survey

References	Methods	Task	Category	Subcat.	Graph modeling	Transformer	Data	Code
[49]	GERL	News	GUTM	SM	Bipartite	Encoder	MSN News	
[50]	TGT	Multi-behavior	GUTM	SM	Bipartite	Encoder	IJCAI, Alibaba (Taobao)	✓
[51]	UGT	Multimodal	GUTM	SM	Bipartite	Encoder	Amazon (Sports, Clothing, Baby)	✓
[52]	KGTE	Paper	GUTM	SM	Heterogeneous	BERT, encoder	Aminer, DBLP	
[53]	PREMIER	Medication	GUTM	EM	Homogeneous	BERT/encoder	MIMIC-III, PRIVATE	
[54]	RecipeRec	Recipe	GUTM	EM	Heterogeneous	Encoder	Data from food.com	✓
[55]	KECRS	Conversational	GUTM	EM	Heterogeneous	Encoder, decoder	REDIAL	
[56]	CGSNet	Session	GUTM	EM	Bipartite	Encoder	Diginetica, Nowplaying, Alibaba (Tmall)	
[57]	MSR	Session	GUTM	EM	Homogeneous	Encoder	Diginetica, Gowalla, LastFM, Yoochoose	
[58]	MBHT	Sequential	GUTM	EM	Hypergraph	Encoder	Alibaba (Taobao), Retailrocket, IJCAI	
[59]	Trans-GAHNet	Medication	GUTM	EM	Homogeneous, bipartite	Encoder	MIMIC-III	
[60]	DSGCL	Sequential	GUTM	EM	Homogeneous	Encoder	Amazon (Sports, Beauty), Reddit	
[61]	G-BERT	Medication	GATM	SIEM	Tree	BERT	MIMIC-III	✓
[62]	GC-SAN	Session	GATM	SIEM	Homogeneous	Encoder	Diginetica, Retailrocket	
[63]	GTN	Social	GATM	SIEM	Bipartite	Encoder	Ciao, Epinions	
[64]	SCG-SPRe	Sequential	GATM	SIEM	Homogeneous	Encoder	Amazon (Baby, Cellphones, Grocery)	
[65]	TMER	Sequential	GATM	SIEM	Heterogeneous	Encoder	Amazon (Auto, Music, Toys)	✓
[24]	GFormer	General	GATM	SIEM	Bipartite	Encoder	Ifashion, LastFM, Yelp	✓
[66]	MGCL	Sequential	GATM	SIEM	Homogeneous, heterogeneous	Encoder	Amazon (Book, CD, Movie)	✓
[67]	MPL-TransKR	General	GATM	SIEM	Heterogeneous	Encoder	Book Crossing, LastFM	
[68]	Hgformer	General	GATM	ESIEM	Bipartite	Encoder	Amazon (Book, CD, Movie), Douban (Book, Movie, Music)	✓
[69]	MIG-GT	Multi-modal	GATM	ESIEM	Bipartite	Encoder	Amazon (Baby, Sports, Clothing)	✓
[70]	SEAGET	POI	GATM	ESIEM	Homogeneous	Encoder	Foursquare	
[71]	GETNext	POI	GATM	ESIEM	Homogeneous	Encoder	Foursquare, Gowalla	✓
[72]	LightGT	Multimodal	GATM	ESIEM	Bipartite	Encoder	Movielens, Kwai, Tiktok	✓
[73]	DA-GCN	Session	GATM	ESIEM	Homogeneous	Encoder	Diginetica, Nowplaying, Alibaba (Tmall)	
[74]	FedTrans4Rec	Sequential	GATM	ESIEM	Heterogeneous	Encoder	Book-Crossing, Last FM, MovieLens	
[75]	MobGT	POI	GATM	ESIEM	Homogeneous	Encoder	Gowalla, Foursquare, Toyota	✓
[76]	KASCRS	Conversational	GATM	ESIEM	Heterogeneous	Encoder, decoder	REDIAL, INSIPIRED	
[26]	Relphormer	General	GATM	ESIEM	Heterogeneous	Encoder	MovieLens	✓
[12]	SIGformer	General	GATM	ESIEM	Bipartite	Encoder	Amazon (CDs, Music), Epinions, KuaiRec, KuaiRand	✓
[77]	DiffGT	General	GATM	ESIEM	Bipartite	Encoder	MovieLens, Foursquare, Yelp	
[78]	G-TransRec	Sequential	GATM	ESIEM	Heterogeneous	Encoder	Alibaba (Taobao), Yelp	
[79]	KGNext	POI	GATM	ESIEM	Heterogeneous	Encoder	Foursquare (NYC, TKY, SIN)	
[80]	MGFormer	General	GTSAM	MAM	Bipartite	Encoder	Alibaba, Amazon (Beauty), Yelp	
[81]	KGNTN	General	GTSAM	MAM	Bipartite	Encoder	Book-crossing, MovieLens, Last-FM	✓
[82]	HUGT	Sequential	GTSAM	MAM	Heterogeneous	Encoder	Movielens, Netflix, Steam, Alibaba (Taobao)	

Table 1 (continued) Summary of the selected works in this survey

References	Methods	Task	Category	Subcat.	Graph modeling	Transformer	Data	Code
[83]	GMT	Click-through-rate	GTSAM	MAM	Heterogeneous	Encoder	WeChat, Alibaba (Tmall)	
[84]	IGT	Session	GTSAM	MAM	Homogeneous	Encoder	Diginetica, Yoochoose	
[85]	FATH	Multi-behavior	GTSAM	MAM	Hypergraph	Encoder	IJCAI	
[86]	TGSRec	Sequential	GTSAM	SMAM	Bipartite	Encoder	Amazon (Baby, Music, Toys, Tools), MovieLens	✓
[87]	SRJGraph	Click-through-rate	GTSAM	SMAM	Heterogeneous	Encoder	Alibaba (Taobao)	
[88]	TrajGraph	POI	GTSAM	SMAM	Bipartite	Encoder	Foursquare (TKY, NYC), Gowalla	
[89]	DGT	Sequential	GTSAM	SMAM	Heterogeneous	Encoder	Amazon (Sport, Movie, CD, Music, Beauty)	✓
[90]	Rankformer	General	GTSAM	MAM	Bipartite	Encoder	Amazon (CD), Yelp, Epinions, Ali-Display	✓
[91]	PMGT	Multimodal	HM	SHM	Homogeneous	Encoder	Amazon (Games, Tools, Toys), MovieLens	✓
[28]	SHT	General	HM	SHM	Hypergraph	Encoder	Gowalla, Alibaba (Tmall), Yelp	✓
[16]	TransGNN	General	HM	SHM	Bipartite	Encoder	Yelp, Alibaba (Tmall), Gowalla, Amazon (Book), MovieLens	✓
[92]	SGTN	Social	HM	SHM	Bipartite	Encoder	Ciao, Epinions	
[93]	L-MTAR	Web-API	HM	SHM	Bipartite	Encoder	ProgrammableWeb	
[94]	KHGT	Multi-behavior	HM	MSTM	Heterogeneous	Encoder	MovieLens, Retail, Yelp	✓
[95]	GraphTR	Multimedia	HM	MSTM	Heterogeneous	Encoder	WeChat Top Stories	✓

one feed-forward network, text tokens through another, yielding a visual and a textual item embedding. The embeddings learned by the transformer are then refined with user-item relationships via a bipartite user-item graph: Each user or item node updates its representation by aggregating 1) the multi-modal embeddings of neighbors and 2) its own multi-modal features. Both the extraction step (the transformer) and the fusion step (the GNN) are trained end-to-end, ensuring that multi-modal signals and graph connectivity reinforce each other. Note that this approach falls within the GUTM category because the graph is never utilized by the transformer itself, but is only integrated through the aggregation function, e.g., $\mathcal{F}$, as illustrated in Fig. 3. On three Amazon datasets (Sports, Clothing, Baby), in terms of Recall@ k /NDCG@ k , UGT outperforms nine multi-modal baselines by up to 13.97%.

Summary. The approaches under this category all rely on a transformer that is fed with only content features – such as images or text – without leveraging any graph information. Graph information is either used to enhance embeddings or constructed from the transformer's output. Specifically, GERL utilizes a GAT layer refining user and news representations via click graphs, while TGT employs a behavior-aware message passing scheme, with messages propagating from item to user side, and generates the user embedding based on a specific behavior type. UGT leverages a shared multi-modal transformer that jointly attends to text and image tokens of items, then injects the combined embeddings into a user-item graph for neighborhood aggregation. In con-

trast to the approaches mentioned above, KGTE uses the transformer's output to construct a heterogeneous knowledge graph, which is then processed using RippleNet over meta-paths. The design of single GUTM is particularly useful when content-side information (e.g., titles, high-quality images, rich attributes) is rich and thus already encodes most of the recommendation logic. In such cases, the transformer captures long-range dependencies and semantic coherence that GNNs often fail to model well due to over-smoothing. However, since the SA mechanism is blind to topological information, popularity bias and cold-start problems might remain unmitigated issues. The learned representations cannot exploit homophily or higher-order proximity, and therefore they might degrade on sparse or evolving graphs.

4.2 Ensemble models

Personalized recommendation for medical prescriptions from electronic records (PREMIER)^[53] produces personalized medications for a patient using two stages. In the first one, given electronic health records (EHRs) data, a transformer or a BERT model learns a vector based on the past medication and the procedure and diagnosis of the current visit. The second stage uses two graphs modeling drug co-occurrences and drug interactions, which are given as input to two separate GAT models. The embeddings produced by the two stages are then combined for the final prediction. PREMIER with BERT outperforms the variant with transformer in the first stage, achieving AUC scores of 0.78 and 0.72 on MIMIC-III^[98] and

PRIVATE data, which are two EHR datasets.

Tian et al.^[54] proposed RecipeRec, a heterogeneous graph learning model for recipe recommendation based on contrastive learning. It utilizes a transformer to encode the interactions between the ingredients within a recipe based on their features. The transformer includes an MHSA mechanism followed by a set attention block^[99], for creating the recipe embeddings. A heterogeneous GNN with a hierarchical attention mechanism produces user and recipe embeddings based on a User-Recipe-Ingredient heterogeneous graph with four types of relations. The final user preference embedding is obtained by combining the GNN-based and the transformer-based representations. RecipeRec reaches NDCG@10 score close to 0.35 on recipe data.

Zhang et al.^[55] proposed *knowledge-enriched conversational recommendation system* (KECRS) that constructs a knowledge graph from a given conversational history by mapping each word to an entity. The embedding learned by a graph convolutional network (GCN)^[7] model feeds a module for computing the probability of recommending an item. Given a conversational context, an encoder-decoder transformer is used for the response generation, in which the encoder encodes the entire conversational history word by word. The output of the transformer decoder is concatenated with the GCN representation for learning the probability distribution over the vocabulary. KE CRS reaches a score of 2.56 distinct 4-gram on recommendation dialogues (REDIAL)^[100], a movie dataset including conversations and utterances related to movies.

Contrastive graph self-attention network (CGSNet)^[56] addresses the session recommendation problem with three graphs: item-level, global-level and session-level, leveraging item connectivity information from different perspectives. Two modules initialized with item features are carried out in parallel: The one employing three GNNs for learning item representations, which are then fused with an attention-based module, and the other being a transformer encoder specifically proposed for sequential recommendation^[101]. The transformer learns local representation reflecting user's short-term preference based on initial item embeddings. The embeddings learned from the two modules are used within a contrastive learning framework. CGSNet achieves MRR@10 scores ranging from 0.08 to 0.19 considering transaction (Diginetica), shopping (Tmall) and music datasets (Nowplaying)^[102].

Multi-dimensional shared representation learning (MSR)^[57] is proposed to tackle session-based recommendation using two separate components: A GAT model learns graph-level item representations based on a session graph constructed through item interactions within a session, while a transformer learns sequential-level item representations from user history sequences. The sequential-level and graph-level item representations are further processed and combined by the recommendation prediction layer. MSR reaches MRR@20 scores between 0.09 and

0.31 on transaction (Diginetica), e-commerce (Yoochoose), music (LastFM) and social (Gowalla) datasets.

Multi-behavior hypergraph-enhanced transformer framework (MBHT)^[58] addresses the multi-behavior sequential recommendation task by predicting the next item of a target action (e.g., purchase) from a user's heterogeneous sequence of page views, cart additions, and purchases. It constructs an item-centric hypergraph whose hyperedges capture both long-range semantic affinities and personalized cross-type behavioral dependencies, refining item representations via hypergraph convolution. These global embeddings are then fused – through an attention-based cross-view aggregator – with the outputs of a low-rank transformer that encodes behavior-aware sequences, while leaving the self-attention mechanism graph-agnostic. The fused vector is used in the final objective, where dot-product scores over all candidate items are passed through a softmax layer to recover the masked target, thereby realizing the recommendation function. Since the graph is processed entirely outside the transformer and only combined downstream, the model falls into the GUTM-EM category in our taxonomy. Empirically, MBHT was evaluated on Taobao, Retailrocket, and IJCAI datasets, yielding gains of up to 52% in NDCG@5 and 27% in HR@10 over various baselines.

Trans-GAHNet^[59] addresses the medication recommendation task by combining a transformer (for patient representation) with multiple graphs (for medication representation). The transformer encodes each patient's longitudinal EHR data into an embedding but ignores structural priors in its self-attention, while three graphs (an EHR graph, a drug-drug interaction (DDI) graph, and a bipartite drug-molecule graph) feed separate GCN modules to capture both safe co-occurrences and potential interaction risks. The final medication combination arises by fusing the transformer's patient embedding and the GCN-derived drug embedding, forming an ensemble design. The authors adopt a transformer to better model irregular visit intervals and long-distance dependencies, thus addressing the cold-start issue and yielding interpretability via the SA module. Experiments on MIMIC-III show F1 scores of 0.6786, plus reduced adverse interactions, thus reconciling personalization with safety.

Dual-view data augmentation at subgraph level and graph contrastive learning for sequential recommendation (DSGCL)^[60] is a sequential recommendation method that combines local sequence modeling with global item relationships. It builds a weighted sequential transition graph to capture multi-step item-to-item dynamics within each user's history, plus an item correlation graph to learn global co-occurrences across all users. User-user and item-item subgraphs are sampled from each global graph and processed by a LightGCN module^[103]. A contrastive loss encourages consistent user representations across the subgraph views. A vanilla transformer encoder receives each user's item list (with positional information) to focus on

local sequential context. Notably, Mu et al.^[60] show that higher-order dynamic information of user-item interactions extracted by subgraph-level extraction mitigates the problem of sparsity of interaction sequential data. Finally, user embeddings from LightGCN and the transformer are fused to predict the next item. DSGCL was evaluated on Amazon and Reddit data, achieving competitive performance in $H@k$, and $NDCG@k$ ($k = \{5, 10\}$).

Summary. Models falling into this sub-category follow an ensemble-like architecture, in which a transformer module captures sequential or semantic patterns, while a separate GNN branch (often LightGCN or GAT) focuses on structural signals derived from the input graph. The prediction of user preferences is obtained by combining the two embeddings or through contrastive alignment (as in CGSNet and DSGCL), which encourages consistency between representations. This design enables the system to be modularized, allowing each component to specialize: The transformer branch models semantic patterns (e.g., long-range dependencies in text data), while the other encodes graph-based information (e.g., cross-user proximity). Such separation is particularly effective in data-sparse or heterogeneous settings, where different types of structure (e.g., conversation, knowledge, medical risk) must be reconciled with user behavior. However, running two full encoders in isolation might significantly increase memory and training requirements. This can also lead to representation mismatch, unstable training, or suboptimal convergence.

4.3 Models with structural input embeddings

For medication recommendation, G-BERT^[61] integrates a tree-structured medical ontology, whose knowledge base is embedded by a GAT model. The diagnosis and medication ontology embeddings are given as input to BERT which is trained on EHRs data from patients having a hospital visit. After obtaining pre-trained visit representations, BERT is fine-tuned for the downstream recommendation task on longitudinal EHRs of patients with multiple visits. G-BERT was evaluated on the MIM-IC-III dataset, achieving F1 score that is equal to 0.61.

Xu et al.^[62] propose *graph contextualized self-attention model* (GC-SAN) for the session recommendation task. Given an item-item session graph based on sequences of anonymous user actions within a session, a GNN model learns graph structured dependencies in the form of node embeddings. Stacked transformer layers, each featuring an SA mechanism followed by a point-wise feed-forward network, encode global dependencies among distant positions, which are instrumental for the final recommendation. GC-SAN achieves $NDCG@10$ scores up to 0.25 on transaction and e-commerce data, like Diginetica and RetailRocket.²

In *graph transformer network* (GTN)^[63], a two-layer

GCN extracts graph topological node embeddings, which are directly used as input tokens by a transformer encoder for learning an enhanced representation for users and items. These embeddings are finally used to predict user/item rating scores. GTN achieves MAE scores between 0.81 and 0.97 on social data.

Substitutable and complementary graph-based sequential product recommendation model (SCG-SPRe)^[64] predicts the products that a user will interact with, in a sequential recommendation scenario. Given two product-product graphs, namely substitutable and complementary³, two separate GNN models capture relation-specific information, thus learning substitutable and complementary product representations. Each representation type is input to a kernel-enhanced transformer, i.e., the SA mechanism is modified for learning temporal and sequential patterns. For capturing temporal patterns between two products, the attention is integrated with different temporal kernel functions that model the dynamics of substitution and complementary products over continuous time. For capturing relations among all interacting products, an SA mechanism learns sequence-aware user representation. The integration of the time-aware and sequence-aware representations is finally used for recommendation. SCG-SPRe achieves $NDCG@10$ between 0.29 and 0.45 on Amazon data^[104].

Temporal meta-path guided explainable recommendation (TMER)^[65] includes an explanation component for sequential recommendation. Given a heterogeneous network of users and items, it initializes user and item representations with DeepWalk^[105], a shallow graph representation learning method. TMER utilizes sequential and non-sequential meta-path instances for characterizing the reasons between each pair of user-item interactions. Paths are regarded as sentences and nodes as tokens in sentences using Word2Vec. TMER employs a MHSA mechanism to learn meta-path-based contexts, then it updates item representations with a two-layer attention mechanism, and uses these embeddings for the final recommendation. TMER scores $NDCG@10$ up to 0.29 on data collected from Amazon.

Li et al.^[24] proposed the *graph transformer* (GFormer) method, which utilizes the transformer to identify self-supervised signals in scenarios with label scarcity. utilizes the transformer to identify self-supervised signals in scenarios with label scarcity. Topological information in the form of node embeddings is first extracted through a GNN, and then is given as input to a MHSA mechanism for rationale discovery, i.e., important edges that guide the model prediction, such as user-item interaction pat-

² <https://www.kaggle.com/retailrocket/e-commerce-dataset>

³ In the e-commerce domain, substitutable relation refers to products that offer similar functionalities. Complementary relations refer to products that are typically used together with another product.

tern highlighting user preferences. The probability of an edge being selected as rationale is computed by aggregating the attention weights learned from each head, while the embeddings learned from the transformer are used to initialize a LightGCN^[103] block, which generates user/item predictions for optimizing a triplet loss function. Alternate blocks of transformer and GNN are also used within the masked autoencoding paradigm^[106], in which rationales are masked from the graph, and an autoencoder model tries to reconstruct them for augmentation. GFormer yields NDCG@10 scores of 0.03, 0.05 and 0.18 on business (Yelp⁴), fashion (Ifashion), and music data (LastFM), respectively.

Multi-view graph contrastive learning (MGCL)^[66] is a contrastive-learning based sequential recommender system in a cross-domain scenario, i.e., rich-source domain information is used for recommending items in a target domain, where source and target domains share the same users. For each user, MGCL learns three GNN-based items embeddings from two domain-specific sequence graphs and a heterogeneous cross-domain graph which aggregates users' behaviors in both domains. A contrastive mechanism applies to these embeddings to transfer the knowledge across domains and extract self-supervised signals. The aggregation of embeddings up to the t -th time step is fed to a transformer encoder to capture user preferences. MGCL achieves NDCG@10 up to 0.36 on Amazon Movie, CD and Book data.

Multi-perspective learning based on transformer knowledge graph enhanced recommendation (MPL-TransKR)^[67] uses a knowledge graph and a transformer encoder for learning high-order neighborhood information between items and users. First, it learns user and item representations from the knowledge graph with an embedding layer based on neighborhood information extraction. Then, stacked transformer layers enhance the user and item representations through an MHSA mechanism followed by a feed-forward neural network. User embeddings are learned by assigning weights to different entities according to user's attention, whereas item embeddings are learned by capturing long distance information between items. These embeddings are used as input to the prediction module. MPL-TransKR achieves AUC scores of 0.85 and 0.87 on Book-Crossing and LastFM datasets, respectively.

Hyperbolic graph transformer (Hgformer)^[68] integrates the local topology of a bipartite interaction graph with its global relational structure, while operating entirely within a Lorentz manifold and maintaining linear attention complexity. Specifically, users and items are first mapped to hyperbolic space; the graph is then processed by a hyperbolic-based GCN, which is a parameter-free convolution that aggregates neighbors through the closed-form hyperbolic centroid. The resulting curvature-aware embeddings enter a hyperbolic cross-attention transformer

that looks at every possible user-item pair. Then, a positive random-feature kernel reparameterizes the similarity matrix, reducing both time and memory complexity to scale linearly with the number of users and items. Because the graph is consumed solely to craft these structural input embeddings and the transformer attends without masks, Hgformer is an SIEM. On 5 Amazon and Douban datasets, Hgformer outperforms different GNN-based and transformer-based competing methods in Recall@ k and NDCG@ k ($k = \{10, 20\}$) with relative improvements ranging from 15% to 19%.

Modality-independent graph neural networks with global transformers (MIG-GT)^[69] is built for multimodal recommendation, where user-item edges come with three information streams: learnable IDs, text, and images. First, each modality is modeled through an interaction graph and processed by an MGDN model – a weight-free, hop-aware diffusion that generalizes LightGCN. It is worth noting that each GNN stops after a modality-specific hop limit. So, for instance, text may look three steps away while images settle after two. The three outputs are then summed and passed to a sampling-based global transformer, which provides global contextual information to the model. For every node, it draws about 10 random peers from the entire graph and runs self-attention only within that small batch. Note that since the sampling procedure is independent of graph topology, MIG-GT is not classified as an SMAM, nor as a hybrid model. Instead, it falls under the SIEM category, as the graph is used solely to construct structural input embeddings, while the transformer operates without edge-based masking. MIG-GT was evaluated on three Amazon data, achieving improvements ranging from 0.21% to about 6% in Recall@20.

Seasonal and active hours guided graph-enhanced transformer (SEAGET)^[70] is designed for next-POI recommendation. It starts from a trajectory-flow graph: a weighted, directed network where nodes are POIs and edge weights count how often two venues are visited consecutively in historical check-ins. This graph is then fed into a two-layer GCN to encode user, POI and seasonality information. In parallel, a graph-based attention module learns a transition-probability map but keeps it outside the sequence model. Check-in sequences (user + POI + time + season) then pass through a transformer encoder that is quadratic only in trajectory length. Note that the SA mechanism is agnostic to graph topological information. After the encoder, the logits are modulated by the pre-learned transition map and a final filter removes venues that are closed at the predicted time. SEAGET was evaluated on Foursquare-NYC, achieving MRR of 0.3413, with small gains on GETNext and other standard baseline approaches.

Summary. SIEMs feed the transformer model with graph-derived embeddings of users and items. Specifically,

⁴ <https://www.yelp.com/dataset>

user and item information is first compressed using methods such as GCNs, DeepWalk, contrastive-learning-based GNNs, or hyperbolic encoders, and is then fed as static tokens to a transformer encoder. In this design, the transformer can devote its full capacity to global contextualization. For instance, Hgformer maps users and items directly into hyperbolic space, while MIG-GT and GTN use a LightGCN and a GCN, respectively. G-BERT grafts a medical ontology tree onto BERT to inject expert semantics. SCG-SPRe splits the graph into substitutable and complementary channels and feeds each through distinct encoders. Overall, decoupling structural encoding from sequence modeling yields modularity – any advances in graph learning (e.g., contrastive objectives, heterophily-aware GNNs) drop in without touching the transformer; the transformer itself is left free to capture global co-occurrence patterns, so SIEMs inherit both local graph inductive bias and long-range attention. However, in all SIEMs, the pre-computed embeddings might be suboptimal when the graph is noisy or sparse. Since attention remains dense, computational cost scales quadratically, and edge attributes or heterogeneity are typically lost unless explicitly re-injected. Furthermore, models with shallow pre-encoders risk under-utilizing the graph, limiting the benefit of this modular design.

4.4 Models with enhanced structural input embeddings

Graph enhanced transformer (GETNext)^[71] addresses the next POI recommendation task as sequence prediction. Given a trajectory flow map modeled as a directed graph, it first learns POI embeddings with a GCN. This embedding is fused with other contextual information including user embedding and time encoding, to capture users' temporal preferences, obtaining a check-in embedding. A transformer encoder, followed by different neural network-based decoders, takes a trajectory as input, represented by a list of the check-in embeddings, to predict next POI. GETNext achieves MRR between 0.21 and 0.36 on social data, like Gowalla and FourSquare.

Light graph transformer (LightGT)^[72] is an approach for multimedia recommendation, where users and items are associated with different modalities (i.e., textual, visual). LightGT employs a lighter version of the transformer encoder, named light self-attention (LSA), which is composed of single-head SA and layer normalization operations. The input to the LSA block is a combination of the modality-specific embedding (extracted by a pre-trained feature extractor), and the user/item structural information is learned through a GCN that serves as positional encoding. By stacking more LSA blocks, LightGT learns an enhanced contextual representation for users and items that is used for modeling the user preference under each modality. LightGT achieves NDCG@10 up to 0.17

on multimedia data (Movielens^[107], Tiktok, Kwai).

Dijkstra self-attention graph convolutional network (DA-GCN)^[73] employs two item-item graphs: Global, based on Dijkstra algorithm across all sessions, and session, based on item relations in the current session. For both graphs, node embeddings are generated by a GNN, then a transformer encoder extracts an enhanced representation for the final prediction layer. To ensure that recent clicked items are given more importance, the transformer is fed with the sum of the session embedding and the encoding of reverse positions of items in a sequence, yielding a sequence embedding. DA-GCN scores MRR@20 up to 0.20 on transaction (Diginetica), music (Nowplaying) and e-commerce (Tmall) data.

Federated sequential recommendation with knowledge-aware transformer (FedTrans4Rec)^[74] addresses the sequential recommendation task based on a federated learning approach where user preferences are modeled in a privacy-aware manner. First, a GCN model captures the local proximity structure of the target entity from a user-item knowledge graph. Then, for modeling sequence relationship, a fusion layer combines the GNN-based embedding with positional information. The obtained embedding is fed into a transformer layer for capturing time-related relations between any two positions in the sequence. The transformer incorporates the replaced-token detection method, which in the sequence processing corrupts the input by replacing tokens with incorrect tokens instead of masking them. The method achieves NDCG@10 from 0.42 to 0.58 on movie (MovieLens), music (LastFM) and book (Book-Crossing) datasets.

Mobility graph transformer (MobGT)^[75] is a POI recommender system designed for urban mobility applications. MobGT constructs three types of homogeneous graphs – spatial, temporal, and categorical to model the overall mobility patterns of users by capturing geographic and transition information among POIs and their categories cooperatively. GNNs are employed to learn POI embeddings from these three global graphs. Additionally, for each user's trajectory, MobGT introduces a local mobility graph to model individual mobility patterns. A mobility graph embedding obtained by combining the global POI embedding and local mobility information is then given as input to a transformer, enriched with a trainable graph positional matrix. The transformer module leverages a specialized spatial-temporal (ST) graph attention mechanism to capture high-order dependencies and explicit correlations between POIs, by harnessing both global and local graph information to model users' mobility patterns. To predict user's next visited location, two different MLPs are used as prediction modules for detecting the next POI and its associated category, respectively. MobGT is evaluated on two social datasets and the Toyota dataset, that is collected from the GPS systems of Toyota vehicles. MobGT outperforms deep learning-based methods and GETNext (discussed above), achieving a

percentage increase of 25.81% in NDCG@5 on the Toyota dataset, and similar performance on Gowalla and Foursquare.

Relational graph transformer for knowledge graph representations (Relphormer)^[26] captures heterogeneous graph information by introducing the Triple2Seq mechanism, which enables dynamically sampling small contextual subgraphs and converting them into sequences while preserving both the structural relationships between entities and the semantic context of the graph. To preserve structural information in the contextualized subgraphs, Relphormer employs a structure-enhanced SA mechanism that integrates topological information by using an attention bias to encode structural information from the adjacency matrix. Specifically, the attention mechanism is not driven by the structure of the graph, but rather an attention bias for each edge is added to the learned attention coefficient. The bias term is computed through a transformation of the power of the normalized adjacency matrix, hence containing structural information between the pair of nodes associated with the attention coefficient. Relphormer is evaluated on the MovieLens dataset against BERT-based methods, demonstrating improvements of 2% in Hits@1 and 1% in MRR compared to the best competing approach.

Sign-aware graph transformer (SIGformer)^[12] addresses the recommendation problem on signed graphs where both positive and negative interactions between users and items can occur. In a recommendation scenario, two users providing the same feedback (positive/negative) to an item suggests that they may have similar preferences. SIGformer leverages a transformer with two key positional encodings: sign-aware spectral encoding (SSE) and sign-aware path encoding (SPE). The first one encodes the graph structure by combining positive and negative subgraphs in a Laplacian matrix. The second one focuses on encoding path patterns between nodes in the signed graph. It assigns unique identifiers to different path types, based on the sequence of positive and negative edges. These paths are then vectorized, with learned parameters for each path type representing node affinities. Then, multiple transformer layers utilize the two positional encodings, which encode the signed graph information, to generate the final prediction. SIGformer is evaluated on the Amazon, Epinions^[108], KuaiRec^[109], and KuaiRand^[110] datasets, achieving improvements ranging from -0.41% to +15.61% in Recall@20 and from -2.77% to +5.87% in NDCG@20, respectively.

Diffusion graph transformer (DiffGT)^[77] leverages a diffusion process, which includes a forward phase and a reverse process on a user-item interactions graph for introducing controlled noise and progressively refining user and item embeddings. It involves two main components: a graph neural encoder and a linear transformer^[111], i.e., an SA module with linear complexity. First, the user/item node representation is calculated through a

LightGCN encoder^[103]. Then, during the forward diffusion process, a directional Gaussian noise is introduced into the graph, aligning with the directional patterns of the data while preserving its inherent structure. This transformation is controlled by a parameter that dictates the noise level. In the reverse diffusion process, the goal is to learn a denoising model that filters out the added noise and recovers the initial distribution. Once the noisy embedding is obtained, a linear transformer denoises the embedding by leveraging a condition vector derived from user's interacted items. Experiments on datasets such as MovieLens-1M, Foursquare, and Yelp demonstrate strong performance of DiffGT when compared against traditional, diffusion-based, GNN-based, and graph-transformer-based (i.e., GFormer) methods, achieving Recall@20 and NDCG@20 scores in the range of 0.071–0.459 and 0.06–0.661, respectively.

G-TransRec^[78] addresses the next-item recommendation task by leveraging two transformer-based sub-modules: local and global information networks. The first one processes user behavior by modeling the sequence of purchased items and time intervals. Specifically, it takes two input sequences: an item sequence and a time interval sequence, which are embedded into fixed-size vectors through one-hot encoding and are given as input to a transformer. The second one incorporates global information by using a graph embedding layer, which generates node embeddings drawing inspiration from the large-scale information network embedding (LINE) model^[112]. This sub-module captures both local and global features of user-item interactions by employing second-order proximity information on a directed user-item graph. Each node has two embeddings: One represents the node itself and another for its contextual relationship with other nodes. The item embeddings are used as the initial input to a transformer layer, generating the global output. The outputs from the local and global information networks are then fused and passed through fully connected layers to produce the final recommendation. G-TransRec is evaluated on the Taobao and Yelp datasets, achieving an average improvement in the range of 20%–60% over all baselines, as measured by NDCG@ k ($k = \{3, 5, 10, 15\}$) and MRR scores.

Knowledge-aware sequential conversational recommender system (KASCRS)^[76] is a conversational recommendation, where entities and user preferences are first extracted from conversational data and used to construct a knowledge graph. In a first pre-processing step, a heterogeneous GNN, i.e., CompGCN^[113], is trained on this graph to learn node embeddings that are used by a transformer-based recommender system. The transformer is then initialized with learned node embeddings combined with positional embeddings to encode the relative position of an element, ensuring that entities in the same position across different sequences shares the same positional information encoding. The transformer is directly trained for the

recommendation task on sequences of entities extracted from dialogues. KASCERS is evaluated on two datasets comprising dialogues, i.e., REDIAL and INSPIRED^[114], outperforming different transformer-based competing methods in Recall@ k ($k = \{1, 10, 50\}$).

Knowledge-graph-enhanced transformer (KGNext)^[79] deals with the next POI recommendation task under uncertain check-in information – data uploaded by users lacks precise location information and is ambiguous – by combining a heterogeneous knowledge graph and a transformer encoder. A GCN processes user–POI–category transitions and interactions to derive structural embeddings that capture both generic movement patterns and personalized preferences. Meanwhile, each user’s trajectory is enriched with time and POI-type embeddings, plus fused user–POI representations, thus enhancing the graph-based input (ESIEM), before being encoded by the transformer. Two auxiliary heads predict POI type and time, and a user-specific attention mechanism adjusts POI/category scores for fine-grained suggestions. Experiments on three Foursquare datasets show that KGNext improves MRR by about 5.5% on average over the strongest competitor (GETNext).

Summary. The architecture of ESIEMs is designed to combine graph-derived embeddings with other contextual signals, such as time, modality, session history, or other external knowledge, before feeding the user/item embeddings into a transformer. This combination step generates context-aware positional embeddings that help the SA module to disambiguate similar graph pattern occurring under different temporal or multimodal conditions. This design enables the transformer to capture fine-grained user preferences and heterogeneous interaction dynamics. GETNext, MobGT, and KGNext exemplify the POI-forecasting branch, progressively enriching GCN-derived spatial encodings with temporal, categorical, and user-specific signals before handing trajectories to vanilla self-attention. FedTrans4Rec marries privacy-preserving GCN embeddings with position encodings, while G-TransRec splits local and global transformers, proving that structural pre-encoders can coexist with order-aware attention without bespoke graph masks. LightGT offers a resource-conscious variant, pairing single-head self-attention with multimodal (image/text) features and GCN positional encodings. Session-centric DA-GCN, dialogue-driven KASCERS, polarity-aware SIGformer, diffusion-oriented DiffGT, and relation-biased Relphormer stretch the paradigm across dual graphs, signed edges, noise injection, and attention biasing, evidencing its breadth and conceptual fragmentation. Overall, ESIEMs are particularly employed to condition transformer’s input tokens with context-aware structural coordinates, letting standard dense attention reason over heterogeneous interactions without customised masking. Such modularity might ease domain transfer – swapping the upstream encoder or side signals suffices for privacy (FedTrans4Rec

or signed-relation (SIGformer) scenarios, avoiding full architectural rewrites. Yet fusion is inherently unstable: naïve concatenation or insufficiently expressive gating mechanisms can induce modality imbalance, whereby dominant signals suppress weaker but informative information, leading to degraded generalization, especially in sparse or long-tail scenarios. Moreover, frozen, graph-only embeddings age poorly as topologies drift, leaving the transformer unable to self-repair stale structure. Finally, also in this case the computation complexity might be expensive because multiple types of encodings (graph-based, positional, semantic) and/or heuristics, and branching fusion paths often coexist within a single pipeline.

4.5 Masked self-attention models

Heterogeneous user graph transformer-message passing (HUGT-MP)^[82] simultaneously captures sequential and graph topology information through a heterogeneous graph where nodes are items and there are two types of edges: sequential and collaborative edges that capture the sequential ordering of interaction and collaborative similarity, respectively. The graph and learnable positional embeddings are fed into a transformer encoder block in which the module of MHSA utilizes a sparse attention matrix to simulate message passing in GNNs. The attention heads are divided into two groups to learn representations from sequential and collaborative edges separately. These embeddings are then fused for obtaining the final user representation. A variant, named HUGT-SE, learns graph structure information extracted by spatial encoding. HUGT-MP and HUGT-SE achieve NDCG@10 scores ranging from 0.10 to 0.25, performing similarly on movie data, like Netflix, Steam, Taobao and MovieLens.

Graph-masked transformer (GMT)^[83] addresses the CTR prediction task. Given an input heterogeneous graph, GMT first collects relevant nodes for the target user-item pair with a neighbor sampling strategy. These nodes are used to construct four types of interaction graphs for neighborhood modeling (e.g., feature-similarity-based subgraph). Stacked layers of graph-masked-transformer, which consists of heterogeneous node feature transformation, masked MHSA and readout modules, are used for learning a representation based on the input node features and the four graphs. Specifically, the graph-masking mechanism enforces the heads to incorporate graph priors dictated by the four interaction graphs. GMT scores AUC values between 0.72 and 0.94 on video (WeChat) and shopping (Tmall) data.

Interval-enhanced graph transformer (IGT)^[84] leverages the graph structure for understanding the dynamic interactions in a session sequence. Given an item-item session graph, where edges are weighted with temporal interval information among items and item features, a transformer encoder learns interactions among items by calculating an attention score for each edge. The recommenda-

tion score is computed by combining the user's short-term preference, corresponding to the representation of the latest clicked item vector, and her/his long-term preference, yielded with a soft-attention mechanism. IGT achieves MRR@10 scores in the range 0.17–0.30 on retail and e-commerce datasets.

Hyper-graph flash attention transformer for multi-behaviour recommendation (FATH)^[85] combines memory-efficient architectures with a flash attention transformer^[115]. FATH leverages a hypergraph where user/item are nodes and hyperedges connect users to items that they have previously interacted with. A hypergraph flash attention mechanism exploits the hypergraph structure by using user-specific query vectors and item-specific key vectors for computing an attention score across the hyperedges. These scores are then used as attention weights to aggregate the corresponding value vectors of items. The attention is performed by ensuring faster training and a reduced memory usage, by minimizing data transfer among high bandwidth memory, static random access memory and back, thus utilizing less memory than the standard transformer. FATH achieves NDCG@10 score of 0.31 on the IJCAI dataset.

Masked graph transformer (MGFormer)^[80] addresses the GNN's limitations in handling large-scale data, by using a kernelized attention mechanism. This mechanism enables computing attention scores between each pair of nodes with linear complexity instead of quadratic complexity. MGFormer operates through three key components. Firstly, it leverages structural encodings to represent the position of each node within the graph. This is crucial in capturing the underlying topology of the user-item interactions. These structural encodings are derived from the singular value decomposition (SVD) of the user-item interaction matrix, which provides a way to embed users and items in a shared latent space that reflects the global graph structure. Secondly, it reduces the complexity of the attention mechanism by replacing the standard attention computation with a dot-product of feature maps, which can be pre-computed and reused across the model. Finally, MGFormer introduces a reweighing mechanism to assign greater attention to influential nodes. Specifically, a learnable sinusoidal degree mask dynamically adjusts the weight of edges based on the degree of the corresponding nodes, hence ensuring that the attention is concentrated on the most important parts of the graph. MGFormer outperforms traditional GNN models across different recommendation datasets such as Yelp. On this dataset, it achieves a Recall@20 of 0.105 and an NDCG@20 of 0.0668, yielding a 7%–8% improvement over mentioned models.

Knowledge enhanced multi-intent transformer network for recommendation (KGTN)^[81] is designed to capture users' multiple intents, i.e., diverse preferences of users reflected in a knowledge graph such as brands and colors, and reduce knowledge noise, i.e., irrelevant or misleading in-

formation within the knowledge graph that can negatively impact the performance. To achieve this dual purpose, KGTN introduces two key components: global intents modeling with graph transformer and knowledge contrastive denoising under intents. The first one captures user intents by employing a transformer which leverages user-item topological information that learns intent-aware representations. This is achieved through predefined intent prototypes that integrate information from user-item interactions and knowledge graph entities, enabling the model to understand the underlying reasons for user preferences. The second component addresses noise in the knowledge graph by using an intent-driven sampling mechanism to select relevant information. It contrasts local and global information in order to reduce the influence of irrelevant or noisy knowledge and ensure more robust representations for both users and items. The final recommendation is achieved by combining the outputs of the two components. KGTN yielded AUC and Recall@20 metrics of 0.9372 and 0.2826, respectively, on the MovieLens dataset, outperforming baseline models in top- k recommendation tasks.

Rankformer^[90] utilizes an SA mechanism guided by the sign relations that items have with respect to user nodes. Specifically, two sign-aware masks are first constructed from the user-item interaction matrix, so that a user node only attends to its positive or negative neighbors. Positive neighbors are items that a user has already interacted with (or, symmetrically, the users who interacted with an item), whereas negative neighbors are the remaining unobserved nodes. These masks act as graph-driven gates that impose a structural prior on each attention head, ensuring that positive and negative feedback signals are handled separately. Rankformer lets the embeddings evolve in alignment with the gradient of Bayesian personalized ranking (BPR)^[116] – a pair-wise loss that maximizes the probability that, for a given user, an observed item is ranked above an unobserved one – by explicitly mirroring a single BPR gradient step, modulated by a learnable step size that guides the update toward improved ranking performance. Its ranking-objective-based design yields systematic improvements on public benchmarks: The model achieves NDCG@20 of 0.065 on Ali-Display, 0.055 on Epinions, 0.083 on Amazon-CDs, and 0.048 on Yelp, beating strong graph baselines such as LightGCN and MGFormer by up to seven percentage points on the same settings.

Summary. MAMs encode graph priors directly into attention masks, limiting each head to focus on specific node subsets. A common design choice is to translate graph edges into sparse or block-diagonal masks that gate every attention head. Here, each head is constrained to attend to a specific subset of nodes, based on graph topological information. The way each method instantiates this principle differs. HUGT splits heads by behavioral channel so that sequential and collaborative edges are disentangled, letting the encoder fuse order-aware and simil-

arity-aware signals only after independent refinement. GMT generalizes this idea to heterogeneous neighborhoods by supplying a distinct mask per interaction graph, forcing the model to reason over multiple relational views in parallel. IGT refines attention weights dynamically with temporal edge scores, thereby turning the mask into a soft routing matrix that adapts to session context rather than remaining static. FATH is more focused on scalability aspect: It pairs hypergraph-derived masks with flash-attention, showing that sparsity alone is not enough unless memory-efficient kernels are co-designed with the masking scheme. MGFormer, KGTM and Rankformer extend basic masking with loss-aware or knowledge-aware updates, showing that masks can be learnable objects whose structure evolves during training instead of being static or hand-crafted.

Overall, the typical sparsity pattern of MAMs follows a message-passing mechanism, where each node attends exclusively to its neighbors. This design efficiently reduces computational costs because it replaces full attention with edge-guided or position-aware interactions, while also injecting inductive biases that align with user-behavior structures or domain-specific semantic information. When the mask is sufficiently sparse, the inference complexity is subquadratic, hence making MAMs attractive for handling large-scale graphs^[18]. Another important aspect is related to interpretability and noise suppression, because non-zero attention paths coincide with explicit graph connections, and therefore potentially spurious long-range dependencies are pruned by construction. However, incorporating sparse matrices into the self-attention mechanism can also hinder performance. The constrained receptive field may fail to capture emergent relations when the graph evolves; static pruning can sever valid long-range dependencies; and sparsity patterns optimized for one domain rarely transfer effectively to others. Moreover, when the mask is dense or multi-view (e.g., GMT), the theoretical efficiency gains may diminish, while dynamic variants (e.g., IGT, Rankformer) introduce additional parameters and potential training instability. Current MAMs assume a relatively clean and complete interaction graph. However, in cold-start or highly sparse settings, hard masking can be detrimental to capturing essential contextual signals.

4.6 Sampled masked self-attention models

Zan et al.^[86] proposed the *temporal graph sequential recommender* (TGSRec) method, which uses a timestamp-weighted bipartite graph and a temporal collaborative transformer (TCT) initialized with temporal embeddings to unify sequential patterns and temporal collaborative signals. For each user, TCT randomly samples a fixed number of item interactions until time t . In the SA module, the query input is from the target node, while key and value correspond to the stacked information of all sampled interactions. The SA mechanism also considers

the impact of a user's interactions via the temporal information on the edges. TGSRec reaches MRR between 0.21 and 0.38 on Amazon and MovieLens datasets.

Search and recommendation joint graph (SRJGraph)^[87] is a method proposed for the CTR problem. SRJGraph constructs a heterogeneous graph with user-item edges, and given a user-item pair, it samples K-order neighbors for both nodes of the pair and enhances user and item representations through a transformer model. Given a node, value and key vectors are based on the sampled high-order neighbors, while the query is based on the representation of the current user/item node. The final embedding is then computed by aggregating the neighbors' representation from the K -th to the 1-st order recursively. This enables the model to explore higher-order information between users and items. SRJGraph achieves AUC score of 0.65 on Taobao data.

TrajGraph^[88] addresses the next-location recommendation task by transforming user mobility data into a temporal bipartite graph (where nodes are users and POIs, and edges reflect time-stamped visits). Notably, the self-attention mechanism focuses on sampled neighbors (instead of all nodes), thus enforcing a masked attention that restricts each node's receptive field. Specifically, a spatial view lists the most recent neighbors (users or POIs) in descending time order, reflecting local/short-term context. A temporal view samples from the latest time window to capture the overall trend leading up to the node's last interaction. Each node in these sampled sequences is augmented with a time-interval encoding. The sequences then feed into a graph transformer block that applies self-attention across only the sampled elements, preventing quadratic costs and leveraging essential neighbors for representation learning. TrajGraph achieved an average MRR increase of 5.21% over three real-world mobility datasets (TKY, NYC, and Gowalla), when compared with other Next POI recommender systems, such as GETNext.

Disentangled graph transformer (DGT)^[89] is a sequential recommendation method which leverages prior knowledge of user-rating and item-rating distributions. Specifically, it constructs a temporal heterogeneous graph of user-item interactions and splits each user's (and item's) history into two subsets (higher VS. lower ratings) to extract disentangled embeddings (preference/disfavor). These embeddings are learned via a transformer where each node (user or item) only attends to its sampled neighbors from the temporal graph (e.g., top-k latest interactions). Furthermore, DGT encodes each edge with rating/time embeddings, then performs a dot-product attention over these neighbor embeddings. For each user, positive-rating neighbors contribute to a preference vector, while negative-rating neighbors feed into a disfavor vector, ensuring disentangled user representations. A similar two-branch scheme is also used to model items. During this attention step, DGT explicitly addresses selec-

tion bias, i.e., the tendency for high or low ratings to appear unevenly because users only choose to rate certain items by correcting the attention weights of neighboring nodes. Finally, user/item embeddings are concatenated and fed into an MLP layer to predict the target rating. Experiments on five Amazon datasets show that DGT achieves competitive performance when compared to classical, GNN-based and transformer-based competing methods.

Summary. SMAMs all augment masking with neighbor sampling techniques. Nevertheless, methods falling in this category differ radically in what gets sampled and why: TGSRec draws a fixed-size window of a user's most recent interactions, fusing sequential order with timestamp-weighted collaborative ties to keep temporally salient evidence in view. However, stochastic sampling introduces variance in the prediction task, i.e., repeated inference passes are required for stable scores. By contrast, SRJGraph emphasizes structural depth by sampling high-order neighbors within a heterogeneous graph. This enhances the semantic richness of user and item representations, but doesn't directly ensure that the computational cost of attention reduces. From this point of view, Traj-Graph's restricted SA module is more scalable because it carefully selects spatial and temporal neighbors, applying a masking mechanism that avoids quadratic complexity and improves context relevance. This might lead to potential useful nodes being omitted. On the other hand, DGT partitions each user-item history into positive-rating and negative-rating groups, samples within each branch, and feeds the two sets into disentangled attention heads that explicitly separate preference from aversion while re-weighting edges to counter selection bias. Here, sampling serves more as a means of bias mitigation than efficiency enhancement.

Overall, injecting a sampling strategy within a graph-driven SA mechanism can help to alleviate memory requirements and enable exploitation of higher-order structure without hand-crafting large static masks or using a full attention mask. Also in this case, the typical sparsity pattern follows a message-passing scheme, where the sampling procedure resembles the approach used in GNNs^[117]. However, this approach might introduce stochastic variability^[118]. Moreover, all the works reviewed in this survey rely on heuristic selection rules across models, which may not always be suitable across all recommendation domains.

4.7 Single hybrid models

Pre-trained multimodal graph transformer (PMGT)^[91] is an unsupervised pre-training strategy for learning item representations in a multimodal recommendation scenario. Given item-item graphs for each mode, an attention-based module aggregates an item's multimodal information based on their contextual neighbors. The multimod-

al representation of each item is summed with a position-id embedding, encoding the order of contextual neighbors, and a role-based embedding, differentiating if a node is target or a contextual neighbor, thus obtaining the initial input embedding for each item. The position-id and role-based embeddings encode structural information of the graph (GATMs). Then, a transformer encoder promoting diversity between the target nodes and their contextual neighbors by ensuring that similar nodes have a lower attention score, is applied for obtaining the final embedding (GTSAM). PMGT obtains NDCG@10 scores ranging from 0.16 to 0.48 on e-commerce and movie data.

Self-supervised hypergraph transformer framework (SHT)^[28] preserves global collaborative relations between user and items through a hypergraph-guided transformer architecture, named hypergraph transformer (HT). SHT learns user and item topological embeddings with a two-layer LightGCN, which is then used to initialize the HT (GATM). Multiple layers of HT are stacked for encoding the global collaborative relations. Each layer conducts hypergraph information propagation by following a message passing scheme (GTSAM). It first computes hyper-edge embeddings with an MHSA mechanism between user/item nodes and hyper-edges. After enhancing the embedding with another GNN module, the process is reversed by propagating the hyper-edge embeddings from hyper-edge to user/item nodes sharing the query, key and values vectors. SHT is tested on 3 datasets spanning different domains, reaching higher Recall@ k and NDCG@ k (with $k = \{20, 40\}$) scores compared to different classes of competing methods on the Yelp, Gowalla and Tmall datasets.

TransGNN^[16] alternates transformers and GNNs to leverage the advantages of both architectures. Specifically, transformers expand the receptive field of GNNs, hence mitigating the over-smoothing phenomena, while GNNs are used to capture crucial structural information. It first samples the most relevant nodes using an attention mechanism that is based on a graph's semantic similarity matrix. Next, the TransGNN module receives input from a positional encoding module, which captures graph topological information like the shortest path, degree, and PageRank-based encodings. This module then alternates between transformers and GNNs in sequence. Interestingly, the transformer layer optimizes its focus on important yet distant information by taking into account the central node and its attention sample, which is based on the aforementioned graph topological information. The attention sample is updated at each step with the new representations with message passing based update strategy. TransGNN is tested on various social datasets, achieving an average performance improvement of about 12% in NDCG@20 across all datasets.

Self-supervised graph transformer network (SGTN)^[92] addresses the social recommendation task with a rating-aware graph transformer for modeling user-item interac-

tions, and a GNN with attention for processing social relationships. First, it decentralizes the initial graph into user-centric interaction subgraphs (user and related items) and item-centric popularity subgraphs (item and its users), while the social domain is captured by a user-centered social subgraph. On the one hand, SGTN introduces a transformer encoder with an edge pipeline mechanism that injects explicit rating embeddings into the self-attention module alongside the implicit pairwise attention scores, thereby fully exploiting rating-based edge features for global user-item context modeling. On the other hand, the GNN aggregator leverages an attention mechanism to account for local social relations and to alleviate data sparsity. Finally, the transformer-derived embeddings are fused with the GNN-derived social embeddings through a three-layer perceptron to produce the final rating prediction. Experiments on Ciao and Epinions report reduced MAE/RMSE scores over some of GNN-based and transformer-based competing methods.

Motif-informed linearizing graph transformer (L-MTAR)^[93] addresses the Web-API recommendation task, i.e., the problem of suggesting the next Web service that a developer should invoke when composing a mashup. It models the system as a directed bipartite invocation graph whose two node types are mashups and APIs. Specifically, an edge records every past call from a mashup to an API, so the graph captures both popularity and invocation order. To capture higher-order structure, the method identifies six small network motifs, i.e., specific four-node patterns that occur significantly more frequently in the bipartite graph than what would be expected by random chance. Each motif induces its own adjacency matrix. The Laplacian eigenvectors of these matrices act as spectral coordinates that quantify how many motif instances connect any pair of nodes and reveal indirect but frequent co-invocation paths that plain edges ignore. These spectral codes are concatenated with a PLM-based semantic embedding (GLM-3^[119]). Then, a motif-aware transformer encoder scans the whole graph: Soft-max is replaced by a non-negative activation that keeps complexity linear, and every attention weight is re-scaled by a function of the key node's motif frequency, giving extra influence to structurally central vertices. Because the graph shapes both the input tokens and the SA weights, the approach is HM. The final embeddings of mashup and API pass through twin multilayer perceptrons (MLPs) and their dot product yields the relevance score. On the ProgrammableWeb benchmark, it achieves HR@10 of 0.931 and NDCG@10 of 0.908.

Summary. HMs utilize a unified transformer-centric framework, where attention weights are simultaneously conditioned on input contextual graph-based information, i.e., $\mathcal{R}$, and graph priors directly injected into the SA mechanism, i.e., $\mathcal{M}$ (Equation (3)). PMGT realizes this by summing multimodal position-id and role embeddings with item-level features while a diversity-oriented mask

suppresses over-attending to near-duplicate neighbors drawn from item-item graphs. SHT stands out for its architectural design, which propagates information via hyperedges within a transformer-guided message-passing loop, embedding high-order connectivity directly into the attention mechanism. TransGNN fuses the shortest-path, degree and PageRank encodings with a dynamically updated attention sample; each layer performs internal GNN propagation inside the self-attention, allowing topological context to be refined before the next pass. L-MTAR linearizes a mashup-API graph by weighting attention with motif frequencies while feeding spectral coordinates as token features, weaving higher-order patterns directly into the kernel without extra stacks. SGTN injects explicit rating embeddings alongside implicit attention scores to model preference intensity in a single encoder, then merges these with social GNN outputs through an MLP.

Notably, the tight integration between the two inductive biases (described by $\mathcal{R}$ and $\mathcal{M}$) enables hybrid models to mitigate limitations faced by purely structural or sequential recommender systems. For instance, SHT and TransGNN alleviate over-smoothing by alternating between structural propagation and attention refinement, while PMGT and L-MTAR tailor attention weights based on graph-derived constraints (e.g., diversity scores or motif frequencies), thus encouraging better disentanglement of semantic and structural information. Such coupling seems particularly effective in sparse or cold-start scenarios, where global attention alone would overfit to popular nodes. However, hybrid systems require careful alignment between message-passing steps and attention mechanisms and often rely on handcrafted graph signals that may not generalize well across domains. Additionally, these models can be difficult to scale due to multi-stage computations and non-standard attention formulations.

4.8 Multi-stack transformer models

Knowledge-enhanced hierarchical graph transformer network (KHGT)^[94] deals with multi-behavior recommendation using two heterogeneous graphs: The one for modeling user-item relations and the other for modeling items' dependencies based on their external knowledge. Each graph is processed by its own transformer stack. The first one employs a GNN-like MHSA, while the second one receives node embeddings computed by a lightweight R-GCN. Also, motivated by the transformer positional embedding, a temporal encoding strategy is utilized both for users and items in order to capture the influences between different types of user-item interactions in a time-aware scenario. Another SA module learns the relevance between different behaviors, yielding a user/item enhanced type-specific behavior representation. KHGT achieves NDCG@10 scores of 0.27, 0.597 and 0.60 on retail, movie and business data.

Graph neural network based tag ranking (GraphTR)^[95] deals with recommendation cases where videos are associated with tags. The problem is modeled with a heterogeneous network with videos, tags, users and media as entities, and 5 types of edges (e.g., video-video, video-user, etc.) for learning from multiple interactions. A transformer is employed in two distinct layers. In the first one, a multi-field transformer operates in parallel with a GraphSage model^[117]. The multi-field transformer utilizes an MHSA mechanism in which neighboring nodes are

separated into four fields according to their type. In the second aggregation stage, given the concatenation of the embedding learned by the GNN model and the multi-field transformer, a transformer encoder learns the final embedding for all types of target nodes. GraphTR is regarded as a hybrid approach since it falls in both GT-SAM and GATM classes, due to the first and second aggregation stages, respectively. Specifically, one transformer stack applies a graph-driven routing mask (multi-field transformer) while a separate stack ingests structur-

Table 2 Cross-model performance comparison. Evaluation measure and category indicate the evaluation measure used by the corresponding method and the category assigned to that method in our proposed taxonomy.

Datasets	Evaluation measure	Methods	Category	Task	Score
Alibaba-Taobao	HR@10	TGT	GUTM	Multi-behavior	0.452
		MBHT	GUTM	Sequential	0.405
		HUGT	GTSAM	Sequential	0.173
		G-TransRec	GATM	Sequential	0.165
		MBHT	GUTM	Sequential	0.257
	NDCG@5	TGT	GUTM	Multi-behavior	0.180
		G-TransRec	GATM	Sequential	0.061
	NDCG@10	TGT	GUTM	Multi-behavior	0.364
		MBHT	GUTM	Sequential	0.283
		HUGT	GTSAM	Sequential	0.105
		G-TransRec	GATM	Sequential	0.083
Alibaba-Tmall	Recall@40	TransGNN	HM	General	0.030
		SHT	HM	General	0.040
	NDCG@20	TransGNN	HM	General	0.100
		SHT	HM	General	0.065
	NDCG@40	TransGNN	HM	General	0.056
		SHT	HM	General	0.035
Amazon-Baby	Recall@10	TGSRec	GTSAM	Sequential	0.224
		MIG-GT	GATM	Multimodal	0.066
		UGT	GUTM	Multimodal	0.060
		TGSRec	GTSAM	Sequential	0.223
	Recall@20	MIG-GT	GATM	Multimodal	0.102
		UGT	GUTM	Multimodal	0.093
		SCG-SPRe	GATM	Sequential	0.292
		TGSRec	GTSAM	Sequential	0.216
	NDCG@10	MIG-GT	GATM	Multimodal	0.036
		UGT	GUTM	Multimodal	0.033
		MIG-GT	GATM	Multimodal	0.045
		UGT	GUTM	Multimodal	0.041
Amazon-Book	Recall@20	Hgformer	GATM	General	0.129
		TransGNN	HM	General	0.080
	NDCG@20	Hgformer	GATM	General	0.068
		TransGNN	HM	General	0.060

Table 2 (continued) Cross-model performance comparison. Evaluation measure and category indicate the evaluation measure used by the corresponding method and the category assigned to that method in our proposed taxonomy.

Datasets	Evaluation measure	Methods	Category	Task	Score
Amazon-CD	Recall@20	Rankformer	GTSAM	General	0.141
		Hgformer	GATM	General	0.140
	NDCG@20	Hgformer	GATM	General	0.083
		Rankformer	GTSAM	General	0.068
Amazon-Clothing	Recall@10	MIG-GT	GATM	Multimodal	0.064
		UGT	GUTM	Multimodal	0.060
	Recall@20	UGT	GUTM	Multimodal	0.093
		MIG-GT	GATM	Multimodal	0.092
	NDCG@10	MIG-GT	GATM	Multimodal	0.042
		UGT	GUTM	Multimodal	0.040
Amazon-Sports	Recall@10	MIG-GT	GATM	Multimodal	0.075
		UGT	GUTM	Multimodal	0.071
	Recall@20	MIG-GT	GATM	Multimodal	0.103
		UGT	GUTM	Multimodal	0.113
	NDCG@10	PREMIER	GUTM	Medication	0.180
		MIG-GT	GATM	Multimodal	0.041
		UGT	GUTM	Multimodal	0.040
		MIG-GT	GATM	Multimodal	0.051
Amazon-Toys	Recall@10	UGT	GUTM	Multimodal	0.048
		TGSRec	GTSAM	Sequential	0.365
	Recall@20	PMGT	HM	Multimodal	0.303
		PMGT	HM	Multimodal	0.403
	NDCG@10	TGSRec	GTSAM	Sequential	0.371
		TGSRec	GTSAM	Recipe	0.180
		MIG-GT	GATM	Sequential	0.365
		TMER	GATM	Sequential	0.327
		PMGT	HM	Multimodal	0.234
		TMER	GATM	Sequential	0.371
Amazon-Tools	Recall@10	PMGT	HM	Multimodal	0.234
		TGSRec	GTSAM	Sequential	0.246
	Recall@20	PMGT	HM	Multimodal	0.236
		PMGT	HM	Multimodal	0.302
	NDCG@10	TGSRec	GTSAM	Sequential	0.256
		TGSRec	GTSAM	Sequential	0.245
Ciao	MAE	PMGT	HM	Multimodal	0.206
		GTN	GATM	Social	0.973
	RMSE	SGTN	HM	Social	0.966
		GTN	GATM	Social	0.712
Diginetica	MRR@10	DA-GCN	GATM	Session	0.764
		GC-SAN	GATM	Session	0.283
		CGSNet	GUTM	Session	0.183
	MRR@20	DA-MSR	GUTM	Session	0.201
		CGSNet	GUTM	Session	0.192

Table 2 (continued) Cross-model performance comparison. Evaluation measure and category indicate the evaluation measure used by the corresponding method and the category assigned to that method in our proposed taxonomy.

Datasets	Evaluation measure	Methods	Category	Task	Score
Epinions	MAE	SGTN	HM	Social	0.791
		GTN	GATM	Social	0.844
	RMSE	GTN	GATM	Social	1.014
		SGTN	HM	Social	1.048
Foursquare-NYC	ACC@1	KGNext	GATM	POI	0.310
		GETNext	GATM	POI	0.243
		TrajGraph	GTSAM	POI	0.240
		TrajGraph	GTSAM	POI	0.602
	ACC@5	KGNext	GATM	POI	0.571
		GETNext	GATM	POI	0.509
		TrajGraph	GTSAM	POI	0.796
		KGNext	GATM	POI	0.685
	ACC@10	GETNext	GATM	POI	0.614
		TrajGraph	GTSAM	POI	0.422
Foursquare-TKY	ACC@1	KGNext	GATM	POI	0.269
		GETNext	GATM	POI	0.225
		MobGT	GATM	POI	0.221
		TrajGraph	GTSAM	POI	0.736
	ACC@5	KGNext	GATM	POI	0.488
		GETNext	GATM	POI	0.442
		MobGT	GATM	POI	0.430
		TrajGraph	GTSAM	POI	0.815
	ACC@10	KGNext	GATM	POI	0.571
		GETNext	GATM	POI	0.529
		MobGT	GATM	POI	0.494
		TrajGraph	GTSAM	POI	0.579
	MRR	KGNext	GATM	POI	0.379
		GETNext	GATM	POI	0.326
		MobGT	GATM	POI	0.316
		TransGNN	HM	General	0.189
Gowalla	Recall@20	SHT	HM	General	0.123
		TransGNN	HM	General	0.265
	Recall@40	SHT	HM	General	0.180
		TransGNN	HM	General	0.160
	NDCG@20	SHT	HM	General	0.073
		TransGNN	HM	General	0.132
	NDCG@40	SHT	HM	General	0.088
		TransGNN	HM	General	0.088

Table 2 (continued) Cross-model performance comparison. Evaluation measure and category indicate the evaluation measure used by the corresponding method and the category assigned to that method in our proposed taxonomy.

Datasets	Evaluation measure	Methods	Category	Task	Score
Last.fm	Recall@10	KG TN	GTSAM	General	0.157
		GFormer	GATM	General	0.210
	Recall@20	KG TN	GTSAM	General	0.311
		GFormer	GATM	General	0.235
	NDCG@10	FedTrans4Rec	GATM	Sequential	0.513
		GFormer	GATM	General	0.183
MIMIC-III	F1	PREMIER	GUTM	Medication	0.681
		Trans-GAHNet	GUTM	Medication	0.678
		G-BERT	GATM	Medication	0.615
	Jaccard	PREMIER	GUTM	Medication	0.527
		Trans-GAHNet	GUTM	Medication	0.678
		G-BERT	GATM	Medication	0.457
	PRAUC	Trans-GAHNet	GUTM	Medication	0.780
		G-BERT	GATM	Medication	0.696
MovieLens	Recall@1	PMGT	HM	Multimodal	0.316
		LightGT	GATM	Multimodal	0.265
		KG TN	GTSAM	General	0.184
		FedTrans4Rec	GATM	Sequential	0.639
	Recall@20	PMGT	HM	Multimodal	0.481
		KHGT	HM	Multi-behavior	0.469
		HUGT	GTSAM	Sequential	0.327
		KHGT	HM	Multi-behavior	0.861
	HR@10	FedTrans4Rec	GATM	Sequential	0.798
		HUGT	GTSAM	Sequential	0.223
		KHGT	HM	Multi-behavior	0.597
		FedTrans4Rec	GATM	Sequential	0.581
	NDCG@10	PMGT	HM	Multimodal	0.481
		LightGT	GATM	Multimodal	0.177
		HUGT	GTSAM	Sequential	0.122
		PMGT	HM	Multimodal	0.493
	NDCG@20	DiffGT	GATM	General	0.326
		TransGNN	HM	General	0.322
Nowplaying	Recall@20	TransGNN	HM	General	0.060
		SHT	HM	General	0.040
	MRR@20	DA-GCN	GATM	Session	0.080
		CGSNet	GUTM	Session	0.080

Table 2 (continued) Cross-model performance comparison. Evaluation measure and category indicate the evaluation measure used by the corresponding method and the category assigned to that method in our proposed taxonomy.

Datasets	Evaluation measure	Methods	Category	Task	Score
Yelp	Recall@20	MGFormer	GTSAM	General	0.105
		Rankformer	GTSAM	General	0.089
		GFormer	GATM	General	0.088
		TransGNN	HM	General	0.082
		DiffGT	GATM	General	0.072
	Recall@40	SHT	HM	General	0.065
		TransGNN	HM	General	0.151
		GFormer	GATM	General	0.133
		SHT	HM	General	0.109
		TransGNN	HM	General	0.073
	NDCG@20	MGFormer	GTSAM	General	0.067
		DiffGT	GATM	General	0.059
		SHT	HM	General	0.065
		Rankformer	GTSAM	General	0.055
		GFormer	GATM	General	0.044
	NDCG@40	TransGNN	HM	General	0.100
		SHT	HM	General	0.071
		GFormer	GATM	General	0.055

al embeddings generated via GraphSAGE, and both outputs are reunited in an additional transformer layer. GraphTR reaches MRR of 0.38 on data from WeChat TopStories.

Summary. MTHMs employ multiple transformer blocks, each wired to a different graph prior, and then combine their output for predicting user preferences. KHGT runs one stack on the user-item interaction graph with a GNN-style masked attention, while a second stack digests an external knowledge graph whose node embeddings are first distilled by a lightweight R-GCN; a cross-behavior attention head finally reconciles the two views, hence strictly utilizing one transformer for each graph inductive bias ($\mathcal{R}$ and $\mathcal{M}$). GraphTR instead pairs a field-aware transformer – whose heads are hard-masked by node type – with a GraphSAGE-derived structural encoder; their concatenated embeddings are passed to a second transformer that re-attends to the mixed signal, hence showing a pipeline where stacks operate sequentially rather than in parallel. Across models, one stack typically emphasizes fine-grained, relation-specific patterns (temporal encodings in KHGT, type-separated neighbors in GraphTR), whereas the other stack integrates broader semantic information.

This separation allows each transformer block to focus on specific inductive biases (guiding each learning phase in a specific way), such as heterogeneous structure, temporal dynamics, or relational semantics. Subsequent stacks then integrate these complementary signals to produce enriched representations tailored to the recommend-

ation task. These models are more expressive than either GATMs or GTSAMs^[46–48]. However, multiple self-attention blocks increase computational requirements and demand careful alignment of hidden sizes and positional encodings. For instance, deployment in latency-critical settings might be problematic as custom masking schemes compromise compatibility with optimized kernels, and the additional transformer stacks increase inference depth.

5 Cross-model comparison

Table 2 presents the performance of the surveyed methods, as reported in their original publications. For each dataset and model, we selected the best value reported in the respective paper. To construct Table 2, we selected the most commonly used datasets across these works (see Fig. 4), which include Amazon (Baby, Books, CD, Clothing), Alibaba (Taobao, Tmall), Foursquare (NYC, TKY), MovieLens, Yelp, and others. Table 2 was organized by pairing each dataset, or domain-level subset, such as Amazon-Baby or Amazon-Tools, with each performance measure reported for that dataset in the works reviewed in this survey. To enable a meaningful cross-model comparison, we report only rows where at least two methods are evaluated on the same dataset using the same performance measure. The columns are organized as follows: datasets (the first column) and evaluation measures (the second column) are listed in alphabetical order. Methods (the third column) within each block are ordered by performance, based on the corresponding evaluation

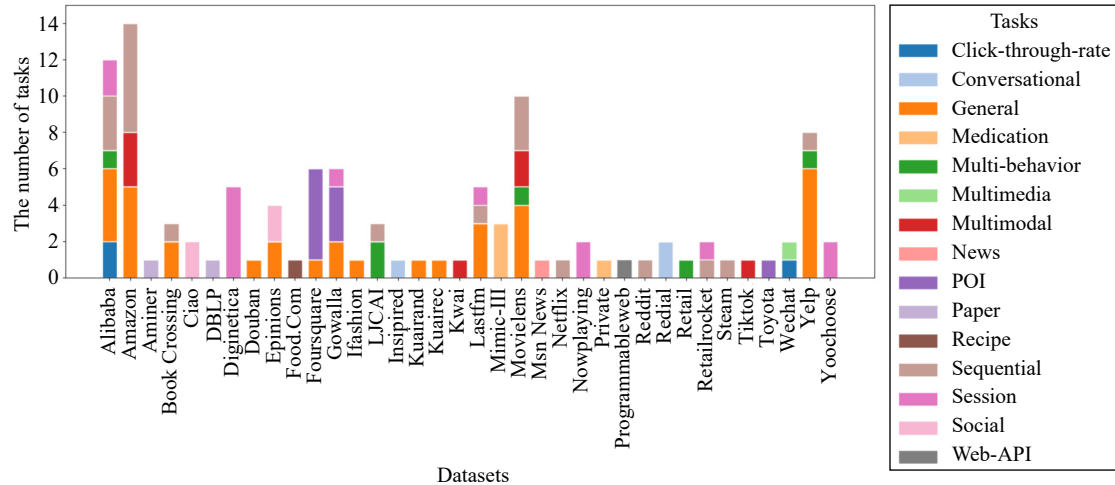


Fig. 4 A stacked bar chart illustrating the number of recommendation tasks per dataset, highlighting the diversity of tasks supported by each dataset. (Colored figures are available in the online version at <https://link.springer.com/journal/11633>)

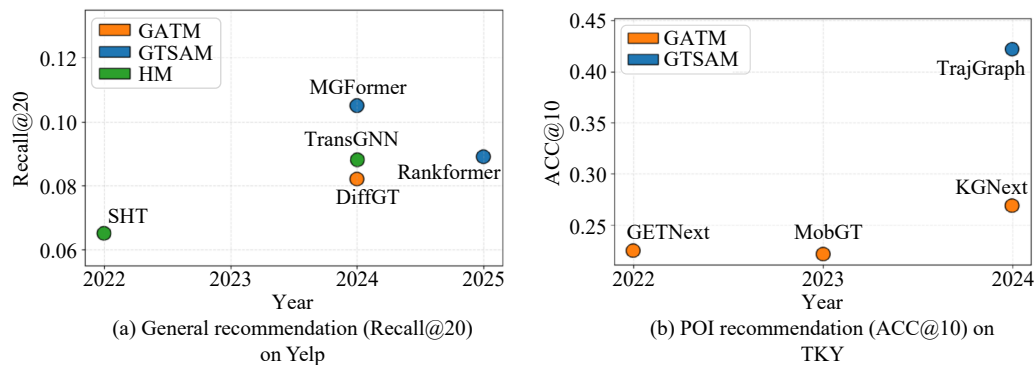


Fig. 5 Progression of GTRS over the years for (a) general and (b) POI recommendation tasks on the Yelp and TKY datasets (Colored figures are available in the online version at <https://link.springer.com/journal/11633>)

measure (the fifth column), with the highest-performing approaches listed first. For each method, we also indicate the specific task for which it was originally designed (the fourth column). This additional dimension enables us to show both when a specialized architecture is employed and how the graph-based recommendation task is modeled, even when operating on the same dataset.

From Table 2, several remarks stand out. First, the performance measures most commonly used by the reviewed works include $NDCG@k$, $Recall@k$, $Hits@k$, and $MRR@k$. In contrast, MAE and RMSE are typically adopted for social recommendation tasks, whereas $ACC@k$ is standard in POI recommendation. Second, TransGNN (HM-SHM) often achieves competitive performance when compared to other general recommenders, particularly when $k > 20$, and consistently outperforms SHT (HM-SM) under the same datasets and evaluation measures. This result suggests that alternating graph-transformer blocks with GNN propagation through a topology-aware attention mechanism, as implemented in TransGNN, may be more effective than embedding and propagating information via hypergraph attention within a single uni-

fied transformer architecture, as done in SHT. Other general recommenders belonging to different categories of our taxonomy, such as Hgformer (GATM-ESIEM) and Rankformer (GTSAM-MAM), achieve competitive performance on data-sparse domains such as Amazon-Book and Amazon-CD, indicating that no single category of models systematically alleviates cold-start issues. Moreover, one of the most widely used datasets for evaluating general recommendation models is Yelp, where Fig. 5(a) illustrates the evolution of these models over the years.

From the perspective of the addressed tasks, further insight emerges. In multimodal recommendation, PMGT (HM-SHM) is the best-performing approaches across several domains: On MovieLens, it is the only multimodal model to exceed $Recall@1 = 0.316$ and $NDCG@20 = 0.493$; on Amazon-Toys/Tools it yields the highest Recall and NDCG scores. Its hierarchical fusion of visual and textual signals via self-attention consistently outperforms MIG-T, UGT, and LightGT.

For POI prediction, trajectory-aware GTSAMs such as TrajGraph are invariably the best-performing approaches on all Foursquare splits – except for $ACC@1$ on

Foursquare-NYC, where KGNext and GETNext (both GATM-ESIEM) prevail – underscoring the importance of spatio-temporal hyper-edges in this task. Fig. 5(b) depicts the evolution of the most widely used models on the Foursquare-TKY dataset.

In the case of medication recommendation, PREMIER achieves better performance than other approaches, such as Trans-GAHNet, which outperforms G-BERT, in terms of F1 and Jaccard scores on the MIMIC III dataset. It is worth noting that both PREMIER and Trans-GAHNet always outperform G-BERT, which may indicate that explicit multi-graph modeling and temporal reasoning yield more effective prescriptions than fine-tuning a BERT-based encoder by leveraging a medical knowledge base. In social recommendation, the standard datasets are Ciao and Epinions, where GTN (GATM-SIEM) and SGTN (HM-SHM) alternately achieve the best MAE and RMSE scores.

Note that no single category of models consistently outperforms the others, confirming that each form of integrating graph structures with transformers offers distinct advantages, whose suitability depends on the application context.

To conclude, we emphasize the lack of standardized evaluation protocols and baselines in the field. Therefore, the comparisons in Table 2 should be interpreted with caution, as each result reflects the specific evaluation protocol used in the original study, including differences in data splits, negative-sampling strategies, the number of test runs, hyperparameter selection, and even the particular cut-off value k used for reporting metrics. Empirical meta-studies on recommender system evaluation have demonstrated that procedural differences alone can alter NDCG@ k or Recall@ k scores by margins larger than the reported algorithmic gains in the literature – occasionally even inverting the relative ranking of methods^[39]. In other words, Table 2 reflects isolated performance tied to specific experimental settings. To transform these results into reliable evidence, the community needs a rigorously defined benchmarking framework: fixed public splits or leave-one-out protocols, standardized negative sampling, and re-ranking pipelines.

6 Main findings, limitations, further perspectives

In this survey, we reviewed recently developed research works that propose GRS approaches using transformers, namely GTRS. Our study has shown that the integration of transformers into GRS has become crucial to many recommendation tasks, since it brings several advantages, such as flexible and improved modeling of complex user-item interactions and the ability to capture long-range dependencies. In the following, we outline the main findings, challenges and limitations of GTRS based on the reviewed works.

6.1 Why leverage transformers for graph-based recommender systems?

The reviewed research works demonstrate that transformers are increasingly applied to a broad spectrum of graph-based recommendation scenarios, in which they generally leverage the MHSA mechanism to capture crucial user-item relationships and filter out less relevant signals.

Typically, transformers are used to address critical challenges in recommendation, such as cold start, data sparsity, and modeling complexities related to long-range dependencies or user-item heterogeneity. Moreover, according to [90], their design is more flexible and better suited to capture the ranking characteristics of graph-based recommendation tasks than standard GNNs.

Transformers provide a flexible, unified framework capable of integrating heterogeneous data sources, such as multiple behavior modalities, content attributes, and negative interactions. This capability enables transformer-based models to capture long-range dependencies within user-item interaction graphs, overcoming limitations inherent in traditional GNNs, such as over-smoothing and bottlenecks due to shallow aggregation layers^[120, 121]. Therefore, transformers can bridge sparse user-item interactions and uncover indirect associations, significantly alleviating data sparsity issues.

Additionally, transformers are used to explicitly model heterogeneous nodes and relations by incorporating specialized positional encodings based on spectral properties, centrality measures, or the shortest-path distances, thus enriching attention mechanisms with structural insights. The use of distinct node-type and edge-type embeddings further distinguishes user-item roles and interaction types. The approach proposed in [93] also shows that transformers are effective in modeling motifs by injecting higher-order sub-graph patterns directly into the SA process. This adaptability facilitates simultaneous modeling of diverse feedback polarities – such as positive and negative interactions within signed graphs – thereby capturing different interaction signals in a unified attention structure. These strategies enable GTRS to integrate auxiliary information and global connectivity, partially mitigating cold-start issues through content or knowledge graph integration and identifying collaborative signals typically overlooked by GNNs.

Moreover, transformers effectively mitigate biases inherent to recommender systems. Popularity bias, characterized by an emphasis on frequently recommended items, can be addressed by using attention reweighting strategies or learnable relative-degree embeddings, reducing excessive focus on popular entities. Similarly, exposure and selection biases arising from skewed interaction data are addressed through self-supervised objectives and debiasing techniques, such as contrastive or generative auxiliary tasks. These approaches generate augmented

training data, enabling models to uncover unbiased interaction patterns and infer latent user preferences for non-exposed items. Lastly, transformers handle temporal biases and evolving user interests by integrating time-sensitive positional encodings or decay factors within the attention mechanism. This approach emphasizes recent interactions, enabling effective adaptation to shifts in user behavior and minimizing reliance on outdated popularity signals.

6.2 Conceptual design of GTRS

Our analysis of the current landscape of GTRS unveils a number of aspects characterizing the conceptual design of the methods. Across GUTMs, transformers predominantly encode external data, focusing on specific recommendation tasks like news, medication, conversational, or paper recommendation without direct graph structural encoding. In ensemble models, the transformer's output is consistently integrated with the output of a GNN model. Single models instead utilize a transformer to learn an initial user/item embedding, which is further enhanced by other graph learning modules or used for constructing a graph. By contrast, GATMs are typically designed with a graph representation learning model followed by a transformer. Additionally, in most of the cases the topological-based node embedding is directly provided in input to the transformer. Interestingly, the work in [68] focused on embedding nodes in hyperbolic space, where cross-attention scores are computed based on hyperbolic distances. Operating entirely in curved space preserves the hierarchical, long-tail structure that is often distorted in Euclidean representations. In ESIEMs, graph structure information is used as positional encoding or integrated with positional encodings for modeling sequence relationship in the user/item interactions history. Note that in GUTMs and GATMs, a node attends to all the other nodes by computing similarity based on the content or structural information, and dealing with at least a quadratic complexity in the number of nodes. By contrast, GTSAMs compute attention weights over the immediate neighbors of each node in the input graph(s), which can model one or more graph priors. The sparse attention mechanism reduces the computational complexity, while leveraging structural priors helps mitigate overfitting phenomena. SMAMs incorporate sampling procedures within SA to enhance the model's expressiveness (e.g., to consider high-order information) rather than to further address computational complexity concerns. This reflects an ongoing effort to balance model complexity with the need to capture rich relational information in recommendation tasks. HMs, on the other hand, aim to combine the advantages of both GTSAM and GATM models. This dual approach can help alleviate over-smoothing, provide resilience in sparse interaction regimes, and supports interpretable recommendations within heterogeneous graphs.

To summarize, the usage of a global attention mechanism with structural biases and structural encodings, consistently emerges as a key strategy for designing GTRS. Specifically, we found that: 1) GUTMs typically utilize transformer for encoding user/item external content information for recommendation in specific domains, such as news, medication, paper, recipe, or conversational domain; 2) GATMs are the prevalent methods for leveraging graph information, in which transformer learns an improved user/item representation; 3) GTSAMs incorporate graph priors useful for recommendation into the SA mechanism to boost the transformer's generalization abilities in encoding user/item interactions and to reduce the computational requirements; 4) The flexibility of the transformer architecture permits its integration with various types of graphs and facilitates the solution of a wide spectrum of recommendation-based tasks. However, the most commonly used component of the transformer is the MHSA module. Nevertheless, the design of GTRS also heavily relies on the specific task to be addressed.

6.3 Task-specific adjustments

GTRS generally requires task-specific adjustments to accommodate the unique characteristics of each recommendation scenario.

In the news recommendation domain, models often integrate external knowledge graphs of news content and entities to enrich user representations. The transformer architecture is adapted to handle multi-source inputs: For example, one component may use SA over a user's recent reading history while another leverages graph propagation of related news topics or entities.

In medication recommendation tasks, GTRS architectures are tailored to incorporate rich medical relational data and safety constraints. The transformer is commonly combined with a medical knowledge graph (encoding relationships among drugs, diseases, symptoms, etc.), so that patient history sequences are augmented with graph-based context. For instance, a patient's sequence of visits or prescriptions can be modeled with SA, while a GNN module infers interactions among co-prescribed drugs or underlying conditions. Architectural adjustments often include mechanisms to recommend a set of medications simultaneously (rather than a single item), which requires modifying the output layer or decoding process for multi-item prediction.

In recipe recommendation tasks, the transformer is typically adapted to handle a heterogeneous recipe graph with culinary information. The transformer component is typically adapted in a hierarchical fashion to handle this multi-modal information. For example, one level of the model might use a content-focused transformer encoder to embed recipe descriptions or ingredient lists, modeling interactions among a recipe's ingredients via the SA mechanism.

In social recommendation scenarios, GTRS approaches can incorporate a social graph alongside the user-item bipartite graph, providing adjacency-based encodings of friend or trust relationships. A key difference is whether the transformer explicitly attends to social edges (injecting edge features into SA) or whether social embeddings simply enter as part of each user's feature vector. Self-attention helps filter extraneous social links while amplifying relevant neighbor relationships.

For POI recommendation, especially next-POI prediction in location-based services, GTRS approaches incorporate spatial and temporal structures into the model. A common adjustment is building dedicated graphs that reflect geographical proximity or mobility patterns – connecting locations that are near each other or frequently visited in succession. The transformer module is then informed by these graphs, for example, by using attention biases or message-passing layers that prioritize geographically close POIs or typical travel transitions. In practice, multiple graph views can be employed (such as a spatial graph of locations, a temporal co-visit graph, or even a graph of POI categories), and the model's SA layers are extended to account for these relational inductive biases. Additional input features like coordinate embeddings, distance-based encodings, or time-interval embeddings are often introduced so that the transformer can learn periodic visitation habits and distance-aware relevance.

In session-based recommendation (where user interactions are grouped into short, anonymous sessions), GTRS approaches are adjusted to capture the complex item relationships within a single session. The sequence of clicked items is typically transformed into a session graph, treating items as nodes with edges denoting their co-occurrence or adjacency in the session's click sequence. Transformers are then modified to leverage this graph structure: For example, SA might be restricted or weighted according to the session graph connectivity so that an item attends more to its immediate neighbors or relevant co-occurring items.

Because session-based scenarios lack long-term user profiles, these models put emphasis on the contextual relationships of items (and possibly the time intervals between interactions) within the session. The net effect is a transformer that excels at short-term pattern recognition, augmented by graph-based relational signals to disambiguate the user's intent within that brief session.

In sequential recommendation for known users (modeling a user's ongoing interaction sequence over time), GTRS approaches combine transformers with graphs that represent higher-order connections in the data. A prevalent approach is to inject knowledge about item-item relationships into the sequential model: For instance, constructing a global co-interaction graph or a knowledge graph of item attributes and using a GNN or attention over this graph to supplement item embeddings. The transformer's architecture is adjusted to fuse two informa-

tion streams, i.e., the chronological sequence of a user's actions and the web of relations linking the user or items to other entities. Moreover, sequential recommenders often distinguish between a user's short-term preferences and long-term profile. Other task-specific adaptations include incorporating a user-item bipartite graph to capture global preference patterns, which is then used to initialize or influence the transformer that focuses on recent behavior. In some cases, hierarchical transformer architectures are introduced (e.g., one layer modeling item transitions within sub-sequences or behaviors and a higher layer aggregating across them) to reflect the multi-faceted nature of user interactions. Across these sequential scenarios, the inclusion of graph structure – being either a social network, a product knowledge graph, or an item co-occurrence graph – helps the transformer overcome data sparsity and sequence noise by providing additional relational clues, thereby yielding more robust next-item predictions that are attuned to both the user's history and the broader connectivity of the ecosystem.

For conversational recommendation, the transformer architecture encodes the conversation context (i.e., multi-turn dialogue) to infer user preferences and generate item-aware responses. Some methods also integrate a knowledge graph to track mentioned entities and relevant facts, but they differ on the transformer's scope – some apply an encoder-decoder transformer for the text-generation side, while others focus on user preference modeling only. Another discrepancy is whether the conversation is represented as a sequence of tokens or mapped onto a structured graph (e.g., each keyword is a node). Despite these variations, the unifying theme is capturing contextual signals via SA and using an external graph for entity disambiguation.

CTR tasks typically deal with sparse and high-dimensional categorical features describing users, items, and the interaction context. Transformers here are used to learn high-order feature interactions. Recurrent patterns include sampling relevant neighbors or constructing local subgraphs for each user-item pair and feeding that to a masked SA block. While nearly all such solutions rely on attention to replace manual feature engineering, they diverge on how they incorporate graph structure: Some only do neighbor sampling, whereas others design specialized mask matrices that reflect the graph's connectivity, ensuring that attention focuses on relevant neighbors.

Finally, in multi-behavior recommendation, GTRS designs commonly preserve multiple interaction types (e.g., views, clicks, purchases) in a multi-relational user-item graph. The transformer is expanded to either capture each behavior type separately or jointly encode them with a specialized attention mechanism that distinguishes among behaviors. This often entails fusing per-behavior attention outputs in a hierarchical manner, allowing the model to highlight the most predictive interaction type for a given recommendation. Besides, some

methods embed the temporal order or recency of different behaviors to discern their relative importance (e.g., a recent purchase VS. a long-ago view). By preserving the structure of each behavior alongside the user-item relationships, these models more precisely infer how diverse interactions evolve into final consumption decisions.

6.4 Datasets used by GTRS

As shown in Fig. 4, some data domains have been considered more frequently than others and have more diversified applications. For example, datasets like Diginetica, MIMIC-III and REDIAL are employed to address specific tasks within the context of graph transformer such as session recommendation, medication recommendation and conversational recommendation, respectively; Foursquare and its subsets are especially used for next POI recommendation. By contrast, Amazon, Alibaba and MovieLens datasets enable the exploration of various GRS tasks, including multi-modal, sequential and multi-behavior approaches. This is because these datasets are rich in information at both the graph level and exogenous textual data. For example, the Amazon dataset comprises multiple sub-datasets related to diverse product categories such as books (e.g., Amazon-Book), electronics, and more. These datasets include reviews, product metadata, and interaction histories, making them highly valuable for recommender systems. Similarly, MovieLens is a widely used dataset for movie recommendation tasks; it contains user ratings for various films, along with additional information about users and movies, such as genres, release dates, and actors.

6.5 Limitations and further perspectives

Despite the significant advancements enabled by GTRS, several key limitations still exist. Scalability remains a critical challenge due to the transformer's dense attention mechanism, which imposes high computational and memory demands, limiting the feasibility of large-scale applications and real-time recommendations. Current mitigation strategies, such as neighbor sampling or truncated exploration, often compromise recommendation performance and training efficiency.

Interpretability is another important limitation, given that the complex interactions between transformer components and GNNs render these models opaque. This lack of transparency hinders user trust, as the specific user-item relationships influencing recommendations are not clearly understood.

Although in most cases transformers are employed to deal with the data-sparsity and cold start issues, these still impact GTRS performance significantly. Sparse interaction signals reduce the reliability of learned embeddings, particularly for new items or users with limited interaction histories. Thus, despite attempts to alleviate

this through self-supervised learning and data augmentation, sparsity in GTRS remains an unresolved core challenge. Furthermore, the theoretical aspects enabling transformers to alleviate this problem are still unclear in most works.

Additionally, most existing models assume a static user-item graph, limiting their adaptability to dynamic scenarios with evolving user interests and continuous data streams. Consequently, maintaining up-to-date recommendations requires resource-intensive retraining, undermining practical applicability in dynamic environments.

Building on the limitations outlined above, the taxonomy proposed in this survey reveals how each GTRS category embodies distinct properties. GUTMs apply dense SA operations over content-based tokens. Since no structural prior restricts the receptive field, their computational complexity is, in principle, quadratic with respect to the number of users and items. However, their training signal remains decoupled from graph noise, making them potentially more robust in cold-start scenarios. At the same time, this architecture limits interpretability, as each token may influence every other, making it difficult to trace the provenance of a recommendation. GATMs inherit the same SA complexity but incorporate graph-derived embeddings, which can reduce popularity bias and better capture long-range dependencies. This added expressiveness, however, comes at the cost of reduced interpretability: Structural signals are embedded into high-dimensional key and value vectors, rather than being made explicit in the attention patterns. GTSAM models, by contrast, are generally more interpretable, as they often employ static, graph-based attention masks that explicitly encode multi-hop neighborhood information. Masking or weighting the attention matrix using input graph edges reduces the quadratic cost to a sparse, block-wise pattern, scaling with average node degree rather than overall graph size. This makes the resulting attention maps easier to inspect for causal insights. However, such methods suffer from reduced coverage – important long-range dependencies may be pruned along with noisy ones – and require mask recomputation when the topology changes, limiting scalability in dynamic settings. SMAMs partially mitigate this limitation by re-drawing neighborhoods dynamically, restoring expressive reach but introducing stochastic variance that can hinder convergence. HMs combine both principles: structural embeddings expand the contextual field beyond what masked heads alone could capture, while graph-guided masks constrain memory usage and expose semantically interpretable routing paths. However, combining these inductive biases increases model depth and parameter count, making training more costly and less stable. Inference latency also grows, particularly when multi-stack architectures transition from sparse to dense attention blocks.

Therefore, each category exhibits its own characterist-

ics, but, as highlighted in Table 2, no category emerges as universally superior.

Addressing these limitations opens several promising research avenues. First, developing scalable architectures with sparse or adaptive attention is paramount for efficient large-scale and real-time recommendations. Another key avenue is harnessing self-supervised learning to mitigate data sparsity by leveraging unlabeled interactions (e.g., contrastive or generative objectives) for richer representations and improved cold-start handling. Furthermore, although LLM-based recommender systems have been explored in the literature^[122–124], their integration with graph structural data is still in its early stages. The main challenges concern designing appropriate graph encoding strategies that enable LLMs to effectively leverage the structural information inherent in graphs^[125–127]. Moreover, real-world graphs are often so large and heterogeneous that they exceed today's context-window limits, making tokenization computationally impracticable^[122]. Nevertheless, in recent years, a body of work has begun to investigate general approaches for integrating graphs with LLMs, by injecting graph topological information, node attributes and relation patterns into the language modeling process. These LLM-driven approaches have shown promising abilities^[122, 125, 127, 128] that could be pivotal for future GTRS in mitigating data sparsity issues and alleviating the cold-start dilemma, e.g., via zero-shot item retrieval^[129].

Finally, ensuring model transparency and fairness is essential for real-world adoption: Incorporating explainability techniques to highlight important graph connections and enforcing fairness constraints to mitigate bias will foster user trust and support equitable recommendations.

6.6 Summary

Overall, we observed that graph transformers offer a powerful framework for integrating various types of data (e.g., user behavior, item attributes, temporal information) and for modeling complex interaction patterns. They excel at capturing long-range dependencies and higher-order connectivity in user-item graphs, which are crucial for accurate recommendations. Furthermore, the flexibility of transformers allows them to handle heterogeneous graphs and incorporate multimodal information, enhancing the personalization and relevance of recommendations. Nevertheless, as shown in Table 1, in several GTRS, the transformer architecture corresponds solely to the encoder network, while the decoder is employed only in conversational recommender systems. By contrast, the modeling of the graph with respect to the task is more heterogeneous. The most commonly utilized graph type is bipartite, followed by heterogeneous graphs, homogeneous graphs, and hypergraphs.

Nonetheless, several limitations arise in current graph transformer based approaches for recommendation. First,

efficient attention models have not yet been thoroughly explored. Indeed, only a few methods tackle complexity concerns by employing lighter SA blocks or memory-efficient transformers. Second, there is a limited effort in addressing the explainability and/or interpretability aspects of recommendation to highlight the most relevant features for suggesting items. Lastly, a number of methods leverage the power of transformers to alleviate the data sparsity issue inherent in recommender systems, e.g., by leveraging transformer within a contrastive learning paradigm for mining self-supervised signals from unlabeled data. Nevertheless, the theoretical aspects enabling transformer to alleviate this problem are still unclear. In fact, as shown in Section 5, there is no single category of models that systematically stands out as the best performing or that consistently alleviates typical recommendation issues, such as the cold-start problem.

This survey has shown that the exploration of transformers within GRS is still in its early stages. There are different directions for further research, such as the development of structural priors that better reflect knowledge about user/item interactions, and the exploration of pre-training and fine-tuning strategy. Indeed, the remarkable success achieved in other domains suggests that there exists significant potential for advancing GRS through pre-trained and fine-tuned transformer models. Moreover, adapting techniques like transfer learning and domain adaptation could help in dealing with data scarcity and cold-start problems in recommendation. Future work might also focus on integrating causal inference methods with graph transformers to address biases and ensure fair recommendations.

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Declarations of conflict of interest

The authors declared that they have no conflicts of interest to this work.

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